

Neurons and neuroglia

The written notes for the neurons and neuroglia walkthrough. Week 4 competencies 7 to 11.

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Organization of the nervous system

- The central nervous system (CNS) is the brain and spinal cord. It is the integrating center: it receives incoming information, combines it with everything else going on, and selects a response.
- The peripheral nervous system (PNS) is all the neural tissue outside the CNS: the cranial nerves, the spinal nerves, and their ganglia. It carries signals between the CNS and the rest of the body, in both directions.
- The sensory division, also called the afferent division, carries information toward the CNS from receptors in the skin, muscles, eyes, ears and internal organs. Most signals from the organs never reach awareness.
- The efferent division carries signals away from the CNS to effectors, which are muscles and glands.

Efferent branch	What it controls	Example
Somatic motor division	Skeletal muscle	Picking up a pencil (voluntary); many reflexes
Autonomic division (visceral nervous system)	Smooth muscle, cardiac muscle, glands, some adipose tissue	Heart rate speeding up (involuntary)

- The autonomic division has two branches that often have opposite effects on the same organ. The organ's activity at any moment reflects the balance between them.
- Sympathetic branch: fight-or-flight. Prepares the body for stress and effort, for example by raising heart rate and sending more blood to working muscles.
- Parasympathetic branch: rest-and-digest. Supports maintenance functions such as digestion, and slows the heart.
- Enteric nervous system: a network of neurons in the wall of the digestive tract that controls how the gut moves and what it secretes. It can detect a stimulus, integrate it and send out a response on its own, and the autonomic division normally adjusts it.

Afferent

Carrying toward the CNS. The sensory division.

Efferent

Carrying away from the CNS, toward effectors.

Effector

A muscle or gland that carries out the response.

Integrating center

The part of a pathway that receives incoming signals, sums them and selects a response. In the nervous system, the CNS.

Functional classes and shapes of neurons

By direction	What it does	Where
Sensory (afferent) neuron	Carries information toward the CNS	Receptor to CNS
Interneuron	Connects other neurons; carries out integration. Roughly 99 percent of neurons are interneurons.	Entirely inside the CNS
Efferent (motor) neuron	Carries signals away from the CNS to effectors	CNS to effector

- A typical neuron has dendrites, short branches that receive signals, and one axon, a long fiber that carries signals away from the cell body.
- Neurons are sorted by shape by counting the processes that leave the cell body.

Shape	Processes	Where it is found
Multipolar	Many dendrites and one axon	Most interneurons of the brain and spinal cord, and efferent neurons such as somatic motor neurons (cell bodies in the ventral horn)
Bipolar	One process on each side of the cell body	Retina and olfactory epithelium, where they carry signals in the pathways for vision and smell
Pseudounipolar	One stalk that splits into a peripheral branch and a central branch	Sensory neurons for touch, pain and stretch; cell bodies in the dorsal root ganglia

- In a pseudounipolar neuron the action potential travels along one continuous axon, from the peripheral branch straight into the central branch, so the cell body is not in the conduction pathway. The cell body sits to one side in a dorsal root ganglion and still makes the proteins the neuron needs.
- Sensory fibers enter the spinal cord through the dorsal root and efferent fibers leave through the ventral root. If one dorsal root is cut, sensation in the area it serves is lost but voluntary movement is kept. Reflexes in that area are lost too, because the afferent half of each reflex pathway has been cut.

Functional regions of a neuron

1. Dendrites receive signals from other cells. One neuron may receive synaptic input from thousands of others. The signals here are graded potentials: small, local, variable in size, and they get smaller as they spread.
 2. The cell body (soma) holds the nucleus and makes nearly all of the proteins the neuron needs. Signals from every dendrite spread into it and add together, which is called summation.
 3. The trigger zone, the axon hillock and the initial segment of the axon, is where the summed graded potentials are integrated: if the sum reaches threshold, an action potential starts here. It has the highest density of voltage-gated sodium channels in the neuron, so it has the lowest threshold and reaches threshold before any other region. In many sensory neurons the trigger zone lies near the receptor endings instead.
 4. The axon carries action potentials. An action potential is all or none: voltage-gated channels regenerate it at full size as it travels, so it arrives at the terminals the same size as it started. A motor axon from the lower spinal cord to a muscle of the foot can be close to a meter long.
 5. The axon terminals release neurotransmitter. When an action potential depolarizes the terminal, voltage-gated calcium channels open, calcium moves into the terminal down its concentration gradient, and the rise in calcium causes vesicles to fuse with the membrane and release neurotransmitter by exocytosis.
- A stronger stimulus does not make a bigger action potential. It makes each sensory neuron fire more often, and it activates more neurons.
 - An input on the cell body next to the trigger zone has a bigger effect there than an identical input at the far tip of a dendrite, because it has less distance over which to get smaller.
 - If the calcium channels at the terminal are blocked, the action potential arrives normally but no calcium enters, so no neurotransmitter is released and the signal stops at the terminal.

Graded potential

A small, local change in membrane potential whose size varies with the strength of the input and which gets smaller with distance.

Action potential

A large, all-or-none signal that travels the length of the axon without getting smaller.

Trigger zone

The axon hillock and initial segment, where the summed graded potentials either reach threshold and start an action potential or do not.

Synapse

The junction between an axon terminal and the next cell: a neuron, a muscle fiber or a gland cell. The gap between them is the synaptic cleft.

Glial cells

Glial cell	Where	Functions
Astrocyte	CNS	Most common CNS glial cell. Regulates the extracellular fluid around neurons: takes up excess potassium after neurons fire, removes neurotransmitter such as glutamate from synapses, and helps deliver nutrients from the blood to neurons. Its end feet release signals that induce and maintain the tight junctions of the blood-brain barrier.
Oligodendrocyte	CNS	Forms myelin. One oligodendrocyte wraps segments of several axons.
Microglia	CNS	Resident immune cells. They survey the surrounding tissue, and when they detect damage, debris or pathogens they become activated and remove them by phagocytosis.
Ependymal cell	CNS	Lines the ventricles and central canal. Specialized ependymal cells produce cerebrospinal fluid; cilia help keep it circulating.
Schwann cell	PNS	Forms myelin. One Schwann cell wraps one segment of one axon. Helps cut peripheral axons regrow.
Satellite cell	PNS	Forms a capsule around neuron cell bodies in ganglia and regulates the fluid around them. Closely related to Schwann cells but does not form myelin.

- Blood-brain barrier: the endothelial cells of brain capillaries are joined by tight junctions. Small lipid-soluble molecules (oxygen, carbon dioxide, alcohol) cross easily; water-soluble molecules (glucose, amino acids) depend on transport proteins. The barrier keeps out many toxins, and also many drugs.
- If astrocytes cannot remove potassium, extracellular potassium rises, the resting membrane potential becomes less negative, and neurons rest closer to threshold. This is one way astrocyte dysfunction is thought to contribute to some seizure disorders.
- Hydrocephalus: if cerebrospinal fluid cannot drain, it accumulates, the ventricles upstream of the block enlarge, and the rising pressure compresses brain tissue. The fluid volume stays constant only when production equals drainage.
- Because one oligodendrocyte myelinates many axons, losing one oligodendrocyte removes myelin from many axons.

When an axon is cut

1. The part of the axon separated from the cell body degenerates, because it no longer receives proteins from the cell body.
2. In the PNS, Schwann cells survive, help clear debris, line up to form a tube and release growth factors, and the axon can regrow along that path, at roughly a millimeter a day.
3. In the CNS, oligodendrocytes and astrocytes release molecules that inhibit regrowth, and astrocytes form a glial scar.

Myelin and what happens when it is lost

- Myelin is the plasma membrane of a glial cell wound around the axon many times. Membrane is mostly lipid and ions cannot cross the lipid bilayer on their own, so myelin acts as electrical insulation.
- Myelin covers the axon in segments. The short uncovered gaps between them are the nodes of Ranvier.

- Voltage-gated sodium channels are densely clustered at the nodes, with very few in the membrane under the myelin.
- The slow step in conduction is the opening of voltage-gated channels that regenerates the action potential, not current flowing along the inside of the axon. Myelin reduces current leak, so current from one node reaches the next node still strong enough to bring it to threshold and regenerate the action potential there.
- The action potential is regenerated only at the nodes, so it effectively jumps from node to node. This is saltatory conduction. More nodes over the same distance would mean more regeneration steps and slower conduction.
- In electrical terms, myelin raises membrane resistance (less current leaks out) and lowers membrane capacitance (less charge is needed to change the voltage). A wider axon also conducts faster, because it has less resistance to current flowing along the inside.

Direction	Motor protein	Cargo
Anterograde: cell body to terminals	Kinesin	Synaptic vesicles, new membrane proteins such as channels, mitochondria
Retrograde: terminals to cell body	Dynein	Worn-out membrane and organelles for recycling, growth factors released by the target cell

Speed	Rate	Carries
Fast transport	Up to about 400 mm per day	Vesicles and mitochondria
Slow transport	About 0.2 to 2.5 mm per day	Cytoskeleton proteins and dissolved enzymes

- At 400 mm per day, fast transport covers a 1 meter axon in about 2.5 days.
- The slow rate of slow transport helps explain why a cut peripheral axon regrows only about a millimeter a day: the new axon is built largely from cytoskeleton proteins that arrive by slow transport.
- If outward transport is blocked at one point, outward cargo accumulates on the cell body side of the block. If inward transport is blocked, returning cargo accumulates on the terminal side. If dynein fails, worn-out material and growth factors no longer return to the cell body, and over time the neuron degenerates.
- Retrograde transport is also a route of infection. Rabies virus is taken up at motor neuron terminals in muscle and carried back along the axon. The cell bodies of these motor neurons are in the spinal cord, so the virus arrives directly in the CNS and then spreads from neuron to neuron to the brain. Tetanus toxin travels the same retrograde route.
- Herpes viruses use both directions: retrograde transport carries them to a sensory neuron cell body, where they remain dormant, and years later anterograde transport carries them back out along the same axon, which is why a cold sore returns in the same spot.

Axon	Conduction speed
Large, myelinated	Up to about 120 m/s
Thin, unmyelinated	About 0.5 to 2 m/s

Demyelination

1. The myelin over a stretch of axon is destroyed and the axon membrane underneath is exposed.
2. Current leaks out across the bare stretch.
3. Because that membrane was covered before, it has very few sodium channels to regenerate the action potential.
4. If the stretch is long enough, the current fades before the next node reaches threshold, and the action potential stops. This is conduction block.

Disease	Myelin attacked	Pattern
Multiple sclerosis	CNS myelin, formed by oligodendrocytes	Symptoms that seem unrelated (vision, sensation, strength, balance) depending on which axons lose myelin. They come and go as areas become inflamed and then partly recover. A rise in body temperature can briefly bring barely conducting axons into conduction block.
Guillain-Barré syndrome	PNS myelin, formed by Schwann cells	In its most common form, an immune attack on peripheral myelin that often follows an infection. Weakness spreads up the legs and reflexes are lost; conduction is slowed but the response size is near normal, so the axons are largely intact. Most people recover over weeks to months because Schwann cells remyelinate axons effectively.

Axonal transport

- Nearly all proteins are made in the cell body. Almost nothing is made in the axon or terminals, so enzymes, membrane and vesicles have to be transported there from the cell body.
- Cargo travels along microtubules that run the length of the axon. Motor proteins carry it step by step and hydrolyze one ATP for each step. If the microtubules are disassembled, transport stops in both directions.

The resting membrane potential

The written notes for the resting membrane potential walkthrough. Week 4 competencies 1 to 5.

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How much of each ion

Ion	Inside (mM)	Outside (mM)
K ⁺	150	5
Na ⁺	15	145
Cl ⁻	10	108
Ca ²⁺	0.0001	1
Large anions (proteins, ATP)	Many	Almost none

- Values vary slightly between textbooks, but the pattern is the same.
- Potassium is the only ion in the table that is more concentrated inside than outside. Sodium, chloride and calcium are all more concentrated outside.
- The large anions are too big to cross the membrane. They are the negative charge left behind when potassium leaves.
- Ions cannot cross the lipid bilayer on their own. They need channels, which are proteins built from the instructions in genes.

How the inside becomes negative

1. The membrane has many potassium leak channels, which are always open.
2. Potassium is more concentrated inside, so it diffuses out through them, down its concentration gradient.
3. Each potassium ion carries a positive charge out and leaves an unpaired negative charge behind, so the inside becomes negative. The separated charge collects in a very thin layer along the membrane, and only a tiny fraction of the cell's potassium has to leave to produce it.
4. The negative inside attracts potassium back in, so the electrical gradient on potassium acts inward. It grows as more potassium leaves.
5. When the electrical gradient is equal and opposite to the concentration gradient, potassium moves in and out at equal rates, with no net movement. That membrane potential is potassium's equilibrium potential, about -90 mV.

6. A few open sodium channels let a small amount of sodium in, carrying positive charge in and making the inside slightly less negative. The resting potential is about -70 mV.

- Permeability is how easily an ion crosses the membrane, and it depends mainly on how many of that ion's channels are open. The ion with the highest permeability has the most influence on the membrane potential. At rest a neuron is roughly 25 times more permeable to potassium than to sodium.
- If every potassium channel were blocked, potassium could no longer carry positive charge out, but sodium would keep leaking in, so the inside would become less negative. The potassium leak channels are the main reason the inside is negative at all.

Resting membrane potential

The voltage across the membrane of a cell at rest, inside relative to outside, about -70 mV in a neuron.

Concentration gradient

The difference in an ion's concentration across the membrane, also called the chemical gradient. It depends only on concentration, not on charge.

Electrical gradient

The effect of the membrane potential on a charged ion. It depends only on the ion's charge and the membrane potential, not on concentration.

Electrochemical gradient

The concentration gradient and the electrical gradient together. It determines the direction of an ion's net movement.

Equilibrium potential

The membrane potential at which the electrical gradient on an ion is equal and opposite to its concentration gradient, so the ion has no net movement.

Permeability

How easily an ion crosses the membrane, set mainly by how many of its channels are open.

The electrochemical gradient

Ion	Concentration gradient	Electrical gradient at -70 mV	Result
K ⁺	Outward	Inward	They oppose each other; a small net outward force
Na ⁺	Inward	Inward	Both act inward; a large net inward force

Ion	Equilibrium potential (about)
K ⁺	-90 mV
Na ⁺	+60 mV
Cl ⁻	-65 mV
Ca ²⁺	+120 mV

- The difference between the membrane potential and an ion's equilibrium potential is the driving force on that ion.
- At -70 mV, potassium is 20 mV from its equilibrium potential and sodium is 130 mV from its, so the driving force on sodium is much larger.
- The membrane potential is measured with a microelectrode inside the cell and a reference electrode outside, and it is always stated as the inside relative to the outside.

Driving force

The difference between the membrane potential and an ion's equilibrium potential.

Which way will it move?

- When an ion's channels open, net movement of that ion drives the membrane potential toward that ion's equilibrium potential.
- When its channels are closed, the ion cannot cross, however large its electrochemical gradient.
- At -70 mV, if sodium channels open, sodium moves in, because its concentration gradient and the electrical gradient both act inward. The membrane depolarizes toward +60 mV.
- At +30 mV, if potassium channels open, potassium moves out with a large driving force: 120 mV from its equilibrium potential, compared with 20 mV at rest. Potassium carries positive charge out and the membrane potential falls. This is how the membrane repolarizes during an action potential.

The Nernst and Goldman equations

- In this course you read and interpret these equations; you do not calculate with them.
- The Nernst equation calculates one ion's equilibrium potential from its concentrations outside and inside the cell. Potassium at 150 mM inside and 5 mM outside gives about -90 mV.
- If outside potassium rises from 5 to 10 mM, potassium's equilibrium potential becomes less negative, about -72 mV, because the concentration gradient driving potassium out is smaller. This is why high blood potassium depolarizes cells.
- The Goldman equation calculates the membrane potential from the concentrations of several ions and the membrane's permeability to each. It works much like a weighted average of the equilibrium potentials, with each ion weighted by its permeability.
- Potassium's permeability is by far the largest, so the resting potential lies near -90 mV, and sodium's small permeability shifts it to about -70 mV.
- Opening more potassium leak channels moves the membrane potential closer to -90 mV (hyperpolarization). Opening many sodium channels makes sodium permeability exceed potassium permeability, so the membrane potential rises toward +60 mV. This rapid depolarization is the rising phase of an action potential.

What the pump does and does not do

- The sodium-potassium pump (Na⁺/K⁺ ATPase) uses energy from ATP to move ions against their concentration gradients: for each ATP it hydrolyzes, it moves 3 Na⁺ out and 2 K⁺ in.
- Its main job is to maintain the concentration gradients over minutes to hours, because sodium leaks in and potassium leaks out continuously.
- Because each cycle moves 3 positive charges out and only 2 in, the pump also makes the inside slightly more negative directly, but only by a few millivolts.
- It does not set the -70 mV resting potential from moment to moment; the leak channels do that. If the pump stops, the membrane potential changes by only a few millivolts at first. Over hours it drifts toward 0 mV as the concentration gradients run down.

Terms for voltage changes

Depolarization

The membrane potential becomes less negative than rest (moves toward zero or above).

Overshoot

The part of an action potential where the inside is positive.

Repolarization

The membrane potential returns toward its resting value, even while the inside is still positive.

Hyperpolarization

The membrane potential becomes more negative than rest.

- These terms describe the direction of the change relative to the resting potential, not whether the inside is positive or negative.
- Channels come in four kinds: leak channels (no gate, always open), voltage-gated, ligand-gated (chemically gated) and mechanically gated. The channel gating walkthrough covers these in detail.

Resting potential and excitability

- A neuron fires an action potential when its membrane is depolarized to threshold, about -55 mV. The difference between the resting potential and threshold is the depolarization needed to fire, and it determines how excitable the cell is.
- A cell that is depolarized is usually more excitable. If a cell stays depolarized for a long time, the inactivation gates of its voltage-gated sodium channels stay closed, fewer channels are available to open, and the cell becomes less excitable.

Condition	Blood K ⁺ (normal 3.5 to 5.0 mM)	K ⁺ equilibrium potential	Resting potential	Effect
Hyperkalemia (missed dialysis)	7.0 mM	About -81 mV	About -65 mV, depolarized	At first closer to threshold; if it lasts, sodium channels stay inactivated and cells can barely fire action potentials. In the heart: slow, disorganized conduction and risk of cardiac arrest.
Hypokalemia (loop diuretic)	2.8 mM	About -105 mV	About -76 mV, hyperpolarized	Farther from threshold, so a larger depolarization is needed to fire: muscle weakness and cramps. Low potassium also makes digoxin toxicity more likely.

- Treating hyperkalemia: IV calcium gluconate protects the heart by raising the threshold of cardiac cells, without changing the potassium level. Insulin with glucose lowers blood potassium by stimulating the sodium-potassium pump, which moves potassium into cells; the potassium is still in the body, so the effect lasts only a few hours. Dialysis is the only one of the three that removes potassium from the body.
- Lidocaine blocks voltage-gated sodium channels, not the leak channels, so the resting potential barely changes. A stimulus still depolarizes the membrane, but sodium cannot pass through the blocked channels, so no action potentials are generated and no pain is felt. A normal resting potential is necessary for a neuron to signal, but it is not sufficient on its own.

Ion channel gating

The written notes for the ion channel gating walkthrough.

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Leak channels and gated channels

- An ion channel is a membrane protein with a water-filled pore that lets specific ions cross the membrane.
- A leak channel has no gate, so its pore is open all the time. Each leak channel is selective: it lets only certain ions through, and those ions move through it continuously, down their concentration and electrical gradients.
- In a resting neuron most leak channels are potassium leak channels. Potassium leaving through them is the main reason the resting membrane potential is negative.
- Most ion channels are gated. Part of the channel protein, the gate, blocks the pore when the channel is closed. When a specific stimulus acts on the channel, it causes a conformational change, a change in the shape of the protein, that moves the gate out of the pore so ions can pass.
- Leak channels set the resting membrane potential. Gated channels change the membrane potential at a specific time, in response to a specific stimulus, and that change is the signal.

Leak channel

A selective channel with no gate, open all the time.

Gated channel

A channel with a gate that opens only in response to a specific stimulus.

Conformational change

A change in the shape of a protein. In a gated channel it moves the gate out of the pore.

What opens a gate

Channel	Opens when	Where you meet it
Voltage-gated	The membrane potential changes. A voltage sensor, a region of the protein with charged amino acids that sits in the electric field across the membrane, moves when the membrane depolarizes, and that causes a conformational change that opens the gate.	The sodium and potassium channels of axons; calcium channels in axon terminals
Ligand-gated (chemically gated)	A specific ligand binds to a binding site on the channel. The binding site fits only molecules with a particular shape and charge, which is why each channel responds to specific ligands. Binding causes a conformational change that opens the gate, and the channel stays open while the ligand is bound.	Acetylcholine receptors at the neuromuscular junction, where a motor neuron meets a skeletal muscle fiber
Mechanically gated	The membrane is stretched, bent or pressed. The deformation changes the shape of the channel protein and opens the gate. No voltage change or ligand is needed.	Touch receptors in the skin, hair cells of the inner ear, stretch receptors in the walls of blood vessels

- Positive feedback in voltage-gated sodium channels: channels open when the membrane depolarizes, sodium enters and depolarizes the membrane further, and that opens more channels. This is why an action potential, once it reaches threshold, always goes to its full size.
- Ligand-gated channels are the basis of chemical synaptic transmission. One cell releases a neurotransmitter, it diffuses across the synaptic cleft, and it binds to ligand-gated channels on the next cell.
- A drug that binds to the acetylcholine binding sites without opening the channels prevents acetylcholine from binding. The channels stay closed, the muscle fiber does not depolarize, no action potential starts in the muscle, and the muscle does not contract: weakness or paralysis. Some paralytic drugs used during surgery work this way.
- In myasthenia gravis, antibodies attack these acetylcholine receptors and reduce their number, which causes muscle weakness that worsens with repeated use.
- Mechanically gated channels carry out sensory transduction for touch, hearing and the sensing of blood pressure: the mechanical stimulus opens the channels, ions cross the membrane, and the membrane potential changes.

Sensory transduction

The conversion of a stimulus, such as pressure, sound vibration or stretch, into an electrical signal.

The sodium channel has two gates

- Activation gate: closed at rest; opens quickly when the membrane depolarizes.
- Inactivation gate: on the cytoplasmic side of the channel. Open at rest; closes more slowly in response to the same depolarization, and then blocks the pore.
- Sodium can pass only when both gates are open.

State	Activation gate	Inactivation gate	Na ⁺ passes?	Can it open now?
Resting (closed)	Closed	Open	No	Yes, if the membrane depolarizes
Open	Open	Open	Yes, sodium enters rapidly	It is open
Inactivated	Open (or closed briefly during repolarization)	Closed, blocking the pore	No	No, however strong the stimulus

The sequence each time the channel opens

1. At the resting potential, about -70 mV, the channel is in the resting state.
2. As the membrane depolarizes, activation gates begin to open. At threshold, about -55 mV, enough sodium channels are open that sodium entering the cell outpaces potassium leaving it, and the depolarization becomes self-reinforcing.
3. Sodium enters rapidly, because its concentration gradient and the electrical gradient both act inward, and the membrane potential rises to about $+30$ mV. This is the rising phase of the action potential.
4. About a millisecond after the activation gate opens, the inactivation gate closes and blocks the pore from the cytoplasmic side. Sodium entry stops automatically, because the same depolarization that opened the channel also closes it.
5. The membrane repolarizes, mainly because potassium leaves the cell.
6. Only then does the channel return to the resting state: the activation gate closes, then the inactivation gate reopens, and the channel can open again.

Inactivation

The inactivation gate blocking the pore. A channel in the resting state can open when the membrane depolarizes; an inactivated channel cannot open, no matter how strong the stimulus.

Absolute refractory period

The time after an action potential when most sodium channels are inactivated, so no stimulus, however strong, can produce another action potential in that region. It limits how many action potentials a neuron can produce per second, and it is why an action potential travels in one direction along an axon.

Return to the resting state

Recovery from inactivation, which happens only after the membrane repolarizes.

The voltage-gated potassium channel

- It opens in response to the same depolarization as the sodium channel, but it has a single gate and it opens more slowly.
- It opens at about the time the sodium channels are inactivating, and potassium leaving the cell repolarizes the membrane.
- It is also slow to close, so potassium keeps leaving after the membrane potential has returned to its resting value. The membrane potential becomes more negative than rest until those channels close.
- After-hyperpolarization, also called the undershoot, is that period below the resting potential. It contributes to the relative refractory period, when a stronger than normal stimulus is needed to fire the neuron, because the membrane is farther from threshold than usual.

After-hyperpolarization

The period after repolarization when the membrane potential is more negative than rest, because voltage-gated potassium channels are slow to close.

Clinical applications

Patient	What happens	Why
Local anesthetic (lidocaine) before a dental procedure	Within a few minutes the gums and lip on that side are numb	Lidocaine binds inside the pore of voltage-gated sodium channels and blocks it, so sodium cannot enter even when the activation gate opens. The axon cannot produce action potentials, so the pain signal is not conducted to the brain. Lidocaine binds most readily to open and inactivated channels, so it blocks frequently firing nerves most effectively, and small pain fibers are blocked at lower concentrations than large motor fibers.
Severe hyperkalemia	Excitable tissue stops responding; life-threatening arrhythmias in the heart	High extracellular potassium keeps the membrane depolarized, so sodium channels stay inactivated and do not return to the resting state.
Hyperkalemic periodic paralysis	Episodes of severe muscle weakness that start during rest after exercise or after eating potassium-rich foods; blood potassium is often high during an episode; strength is normal between episodes	An inherited mutation in the skeletal muscle sodium channel impairs inactivation. Channels that do not inactivate completely keep letting sodium in and hold the membrane depolarized, which keeps most sodium channels, including the normal ones, inactivated. The muscle fiber cannot produce action potentials, so it cannot contract. Sustained depolarization stops activity rather than increasing it.

Graded potentials

The written notes for the graded potentials walkthrough. Week 4 competencies 12 and 14.

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Two kinds of electrical signal

	Graded potential	Action potential
Where	Input zone: dendrites and cell body	Starts at the trigger zone, travels the axon
Size	Varies with the strength of the stimulus	Always the same size (all or none)
Distance	Short; gets smaller with distance	Long; does not get smaller
Threshold	None	Generated only if the trigger zone reaches threshold
Channels	Chemically (ligand) gated or mechanically gated	Voltage-gated sodium and potassium

Graded potentials on a dendrite

1. A stimulus opens gated channels: a neurotransmitter binds to chemically gated channels at a synapse, or stretch opens mechanically gated channels in a sensory ending.
2. Ions cross, and the membrane potential moves toward the equilibrium potential of the ion whose permeability increased.
3. Sodium entering makes the inside less negative: a depolarizing graded potential. At a synapse this is an excitatory postsynaptic potential (EPSP), because it brings the neuron closer to firing an action potential.
4. Potassium leaving makes the inside more negative than rest: a hyperpolarizing graded potential. At a synapse this is an inhibitory postsynaptic potential (IPSP), because it moves the membrane potential farther from threshold and makes the neuron less likely to fire.
5. The channels close and the membrane returns to rest. A graded potential lasts from a few milliseconds to tens of milliseconds.
 - A stronger stimulus opens more channels or holds them open longer, so more ions cross and the change in voltage is larger. That is what graded means: the amplitude varies with the strength of the stimulus.
 - Example amplitudes recorded at one site: weak stimulus 3 mV, medium 7 mV, strong 12 mV; a stimulus that opens potassium channels, about 8 mV below rest.
 - In a sensory receptor, a graded potential produced by a stimulus such as stretch is called a receptor potential.

Graded potential

A small change in membrane potential confined to one region of the membrane, whose amplitude varies with the strength of the stimulus.

EPSP (excitatory postsynaptic potential)

A depolarizing graded potential produced by a neurotransmitter at a synapse.

IPSP (inhibitory postsynaptic potential)

A hyperpolarizing graded potential produced by a neurotransmitter at a synapse.

Receptor potential

A graded potential produced in a sensory receptor by a stimulus such as stretch.

Why graded potentials fade

- Local current flow: the positive charge that enters spreads along the inside of the dendrite in both directions, away from where it entered.
- Current leak: some of that charge leaves across the membrane through the open leak channels along the dendrite, the same leak channels that set the resting potential.
- Cytoplasmic resistance: the cytoplasm, which contains many proteins and other molecules, resists the flow of charge through it.
- Because of current leak and cytoplasmic resistance, the graded potential is smaller at every point farther from where it started. This loss of amplitude with distance is decremental conduction. Example: 20 mV at point 1 next to the open channels, then 12, 7 and 4 mV at points farther along.
- A wider dendrite has a larger cross section for charge to flow through, so its cytoplasmic resistance is much lower. Current leak does not decrease; a wider dendrite has a little more membrane, and so more leak channels, along each length. Cytoplasmic resistance falls much more than leak rises, so the graded potential spreads farther.

Local current flow

Movement of charge through the cytoplasm away from where ions entered.

Current leak

Loss of charge across the membrane through open leak channels.

Cytoplasmic resistance

The cytoplasm's resistance to the flow of charge; lower in a wider process.

Decremental conduction

The decrease in a graded potential's amplitude with distance from where it started.

What reaches the trigger zone

- The trigger zone is the axon hillock and initial segment. An action potential is generated there only if the membrane potential at the trigger zone reaches threshold, about -55 mV. From rest at -70 mV that takes about 15 mV of depolarization.
- The outcome depends on the amplitude of the graded potential when it arrives at the trigger zone, not its amplitude where it started.
- Graded potentials that overlap in time add together: summation. EPSPs move the membrane potential toward threshold and IPSPs move it away. An action potential is generated only if the net depolarization reaches threshold.
- Spatial summation: graded potentials from different synapses that are active at the same time add together.
- Temporal summation: one synapse is activated again before its previous graded potential has faded, and the two add together.
- Real graded potentials that overlap add up to a little less than their simple sum, but the reasoning is the same.

Inputs reaching the trigger zone	Arithmetic	Voltage	Result
One EPSP, 8 mV (20 mV at the synapse)	$-70 + 8$	-62 mV	Subthreshold, no action potential
Two EPSPs at the same time, 8 mV each	$-70 + 8 + 8$	-54 mV	Suprathreshold, an action potential is generated
Two EPSPs of 8 mV plus one IPSP of 5 mV	$-70 + 8 + 8 - 5$	-59 mV	Subthreshold, no action potential

Threshold

The membrane potential at the trigger zone, about -55 mV, at which an action potential is generated.

Subthreshold

Too small to bring the trigger zone to threshold.

Suprathreshold

Large enough to bring the trigger zone to threshold.

Summation

The adding together of graded potentials that overlap in time.

Spatial summation

Summation of graded potentials from different synapses active at the same time.

Temporal summation

Summation of graded potentials from one synapse activated again before its previous graded potential has faded.

Patient: the pufferfish meal

- Tetrodotoxin blocks voltage-gated sodium channels from the outside of the cell. It does not block chemically gated or mechanically gated channels.
- His resting potential is normal (-70 mV). A weak stimulus still produces a small graded potential, about 5 mV, that fades.

- A strong stimulus produces about 20 mV, which goes past threshold, but no action potential appears, because the action potential depends on the voltage-gated sodium channels that the toxin blocks.
- His neurons can still produce graded potentials in response to input, but they cannot generate action potentials to carry signals along their axons. The action potential walkthrough finishes this case.

The action potential

The written notes for the action potential walkthrough. Week 4 competencies 13 to 16.

To print: these notes as a two-column PDF. Already printed the Week 4 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

The key numbers

Point	Voltage	What sets it
Rest	-70 mV	Mostly the potassium leak (resting potential walkthrough)
Threshold	about -55 mV	The inward sodium current first exceeds the outward potassium current
Peak	about +30 mV	Sodium inactivation gates close and voltage-gated potassium channels open
After-hyperpolarization	about -80 mV	Voltage-gated potassium channels are slow to close
Sodium equilibrium potential	+60 mV	The value the rising phase moves toward
Potassium equilibrium potential	-90 mV	The value the after-hyperpolarization moves toward

The whole action potential, from threshold back to rest, lasts a few milliseconds.

The two voltage-gated channels

Channel	Gates	At rest	When depolarized
Voltage-gated Na ⁺ channel	Activation gate, and inactivation gate on the cytoplasmic side	Activation gate closed, inactivation gate open	Activation gate opens quickly; inactivation gate closes a little later and stays closed until the membrane repolarizes
Voltage-gated K ⁺ channel	One gate	Closed	Opens slowly, at about the time the sodium channels inactivate, and closes slowly

- Both channels are opened by depolarization. The difference that shapes the action potential is timing: the sodium channel responds quickly and the potassium channel slowly.
- A stimulus can open activation gates, but no stimulus of any size can open an inactivation gate. It reopens only after the membrane repolarizes.

Reaching threshold

- A graded potential depolarizes the trigger zone, the axon hillock and first part of the axon, and opens some voltage-gated sodium channels.
- Below threshold (example: a 10 mV graded potential brings the membrane to -60 mV), the inward sodium current is smaller than the outward potassium current through the leak channels. More positive charge leaves than enters, so the membrane returns to -70 mV and no action potential occurs.
- At threshold (a 15 mV graded potential brings the membrane to about -55 mV), the inward sodium current equals, then exceeds, the outward potassium current.
- Threshold is defined by this balance between two currents, so its exact value can vary from neuron to neuron.

1. Sodium enters and the membrane depolarizes.
2. Depolarization opens more voltage-gated sodium channels.
3. More sodium enters, and the membrane depolarizes further.
4. Back to step 2. This cycle is positive feedback. Once it starts, the action potential no longer depends on the stimulus.

The phases

Phase	Voltage	Na ⁺ channels	K ⁺ channels	Ion movement
Rest	-70 mV	Resting (activation gate closed)	Closed	Leak channels only
Rising phase (depolarization)	-55 to about +30 mV	Open	Starting to open	Na ⁺ enters rapidly
Peak (overshoot above 0 mV)	about +30 mV	Inactivation gates close	Open	Na ⁺ entry stops; K ⁺ starts to leave
Falling phase (repolarization)	+30 back to -70 mV	Inactivated	Open	K ⁺ leaves
After-hyperpolarization	down to about -80 mV, then back to rest	Returning to the resting state: activation gate closes, inactivation gate opens	Still open, closing slowly	Extra K ⁺ leaves, moving the membrane potential toward -90 mV

- The rising phase moves toward the sodium equilibrium potential (+60 mV) because the membrane is now far more permeable to sodium than to anything else. This is the same rule as for the resting potential: the membrane potential moves toward the equilibrium potential of the ion the membrane is most permeable to.

- The voltage peaks near +30 mV, well short of +60 mV, because the sodium inactivation gates close and the voltage-gated potassium channels open.
- Potassium leaving through voltage-gated channels repolarizes the membrane. The sodium-potassium pump does not.
- The after-hyperpolarization happens because the voltage-gated potassium channels are slow to close, so for a few milliseconds the membrane is even more permeable to potassium than at rest and the membrane potential moves closer to the potassium equilibrium potential.

Threshold

The membrane potential at which the inward sodium current first exceeds the outward potassium current and the action potential begins; about -55 mV here.

Positive feedback

A cycle in which the result of a process strengthens the process itself: depolarization opens sodium channels, and the sodium that enters causes more depolarization.

Rising phase (depolarization)

The rapid rise from threshold to the peak, carried by sodium entering the cell.

Overshoot

The part of the action potential above 0 mV, when the inside is positive.

Falling phase (repolarization)

The return from the peak toward rest, carried by potassium leaving the cell.

After-hyperpolarization

The period after repolarization when the membrane potential is below the resting potential, caused by voltage-gated potassium channels that are slow to close.

How many ions move

- Only a tiny fraction of the ions cross the membrane during one action potential. Afterward, sodium inside is still about 15 mM and potassium inside is still about 150 mM.
- The concentration gradients remain after an action potential, so a neuron can fire again right away without waiting for the pump.
- The pump keeps the concentrations steady over the long term, across thousands of action potentials. It does not repolarize the membrane after each action potential.

All or none

- Below threshold there is no action potential. At or above threshold there is a full-sized action potential every time.
- Example: graded potentials of 5 and 10 mV stay below threshold. Graded potentials of 16 and 30 mV both produce action potentials that peak at about +30 mV.
- Once threshold is reached, the size of the action potential is set by the voltage-gated channels and the sodium equilibrium potential, not by the stimulus. A larger graded potential only brings the membrane to threshold a little sooner.

All or none

An action potential either occurs at full size or does not occur; its size does not depend on the size of the stimulus.

Coding stimulus strength

- Because every action potential is the same size, the size of an action potential cannot encode stimulus strength.
- Frequency coding: a stronger stimulus makes a single neuron fire more often. A larger graded potential at the trigger zone brings the membrane back to threshold sooner after each action potential. Example in the walkthrough: 2, 5 and 9 action potentials in the same period for weak, medium and strong stimuli.
- Population coding: a stronger stimulus activates more receptors, so more neurons fire. Example: the strong stimulus has four neurons firing instead of one.
- A single axon can use only frequency coding.

Frequency coding

Encoding stimulus strength by how often a neuron fires.

Population coding

Encoding stimulus strength by how many neurons fire.

Refractory periods

Period	When	Channel state	Can a second action potential start?
Absolute refractory period	From the start of the action potential until the sodium channels begin to return to the resting state	Na ⁺ inactivation gates closed	No, no matter how strong the stimulus
Relative refractory period	During the after-hyperpolarization	Some Na ⁺ channels have returned to the resting state; extra K ⁺ channels still open	Yes, but only with a stronger stimulus than usual, and the second action potential is often a little smaller

- Example: during the after-hyperpolarization the membrane is near -78 mV. The same 16 mV stimulus that fired the first action potential brings it only to about -62 mV, below threshold.
- The absolute refractory period lasts about 1 to 2 ms, which sets an upper limit on firing rate: $1,000 \text{ ms} \div 2 \text{ ms} = 500 \text{ per second}$, or $1,000 \text{ ms} \div 1 \text{ ms} = 1,000 \text{ per second}$. Most neurons fire well below this.
- The refractory period also keeps the action potential traveling in one direction along the axon (conduction walkthrough).

Absolute refractory period

The time during which no stimulus of any size can start a second action potential, because the sodium inactivation gates are closed.

Relative refractory period

The time that follows, during the after-hyperpolarization, when a second action potential can start but only with a stronger stimulus than usual.

Patient: a drug that blocks potassium channels

- Dalfampridine (4-aminopyridine) blocks voltage-gated potassium channels. It is used to help people with multiple sclerosis walk.

- In multiple sclerosis, axons lose myelin, which exposes potassium channels that are normally covered by it.
- With the voltage-gated potassium channels blocked, the rising phase and peak barely change, because they depend on sodium. The falling phase is much slower; the membrane still repolarizes as the sodium channels inactivate and potassium leaves through the leak channels. The action potential becomes wider, and the after-hyperpolarization mostly disappears.
- A wider action potential delivers more charge to the next segment of axon, which helps the signal cross a segment that has lost its myelin.
- Blocking these channels throughout the brain can also make neurons more excitable, so the dose is kept low: too much can cause seizures.

How action potentials carry information

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 15 TO 18 OF WEEK 4

22 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

An action potential is the same size every time a neuron fires one. That raises three questions, and they are the plan for this lecture. How can a signal that never changes size tell the brain that one touch is light and another is hard? How does it travel a meter along an axon without fading? And what decides how fast it gets there? We answer the first with the refractory periods and frequency coding, the second with local current, and the third with axon diameter and myelin. Then we change the conditions and predict what happens.

Here is the problem we solve by the end. A dentist injects lidocaine beside the nerve that serves the lower teeth. Within a few minutes the lip and the teeth on that side are numb, and the drill causes no pain. The patient can still feel the dentist pushing on the tooth. A few hours later all feeling is back. The drug never reached the pain receptors in the tooth, and they still respond to the drill.

You need three facts to make a first prediction. Lidocaine blocks voltage gated Na^+ channels. It stays close to where it was injected, partway between the tooth and the brainstem. And the endings of pain receptors respond to the drill with channels that are not voltage gated Na^+ channels. So graded potentials still form in the tooth, and action potentials still start in the axon near the tooth. They stop in the stretch of nerve beside the injection, and nothing reaches the brainstem. Pain is felt only when action potentials reach the brain, and one blocked stretch is enough to stop them. The harder question is why pressure still gets through, when the axons that carry it lie in the same nerve and the same drug. Write down a guess before you read on.

See it move: race two axons

SLIDE 15 OF THE DECK

This is a working model of two axons that are stimulated at the same moment. The clock shows real milliseconds. You can choose which two axons to compare, change the diameter of axon A and take its myelin off, change the middle third of both axons, and change how often stimuli arrive at the trigger zone. The strip at the bottom shows which stimuli fire an action potential and which ones land inside the absolute refractory period. Come back to it as you read the sections below.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-ap-conduction-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-ap-conduction-notes.html)

THREE THINGS TO TRY

- 1 Choose the motor fiber and the slow pain fiber and watch the clock. Axon A arrives in 11 ms and axon B in 1 s. Then take the myelin off axon A. It is still 15 times wider than axon B, and it now conducts at only 3.9 m/s. Compare that with worked problem 2.
- 2 Put the myelin back on axon A and press Remove the myelin, which strips only the middle third. Read the new arrival time. Then press Remove the myelin and warm the body, and last press Apply local anesthetic. Say in your own words why the action potential stops in each case.
- 3 Slide the stimuli up from 100 to 500 per second, and then to 800. Read how many action potentials are fired each time, and compare the ceiling with worked problem 1. Then shorten the absolute refractory period to 1 ms and read the ceiling again.

The refractory periods

SLIDES 3 TO 6 OF THE DECK

Start with what you already have from the last lecture. The voltage gated Na^+ channel has two gates, an activation gate and an inactivation gate, and so it has three states. When it is closed and ready, the activation gate is closed and the inactivation gate is open, and a depolarization to threshold will open it. When it is open, both gates are open, for less than a millisecond. Then the inactivation gate closes and the channel is inactivated. An inactivated channel cannot open, however strongly the membrane is depolarized. It resets to closed and ready only after the membrane has repolarized. Everything in this lecture follows from that one fact.

A refractory period is a stretch of time after an action potential during which that patch of membrane is harder, or impossible, to fire again. The absolute refractory period starts when the membrane reaches threshold and lasts about 1 to 2 ms, which is most of the action potential. During it the Na^+ channels are open at first and then inactivated. No stimulus of any size can start a

second action potential, because a stimulus works by opening closed channels, and there are none.

The relative refractory period follows and lasts a few more milliseconds. Some Na^+ channels have reset and some have not. The voltage gated K^+ channels are still open, so K^+ is still leaving, and the membrane is hyperpolarized, farther from threshold than usual. Two things now work against a stimulus: open K^+ channels let positive charge leave as fast as the stimulus brings it in, and the membrane has farther to go. A second action potential is possible, but it takes a larger than normal stimulus, and it may be smaller than normal because only the channels that have reset can take part. An everyday comparison is a toilet that has just been flushed. While the tank is empty, pressing the handle does nothing, however hard you press. While it is refilling you can flush again, but the flush is weaker.

Refractory periods do three jobs. They keep action potentials separate, so they never overlap or add together the way graded potentials do. They set a maximum firing frequency. And they make conduction one way, because the membrane just behind a traveling action potential is refractory.

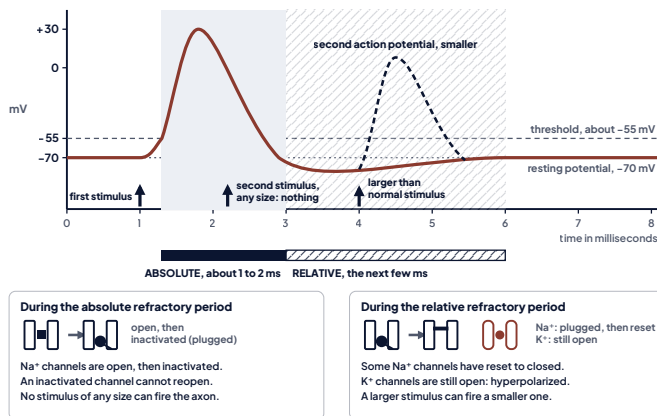


Figure 1. The two refractory periods, with the state of the channels during each one and the result of a second stimulus.

WORKED PROBLEM 1

An axon has an absolute refractory period of 2 ms. What is the largest number of action potentials it can send in one second? What if the refractory period were 1 ms?

maximum frequency = $1 \div \text{absolute refractory period}$, with the period in seconds.

THE WORK

- Put the refractory period into seconds. 2 ms = **0.002 s**.
- One action potential needs 0.002 s before the next one can start, so the most that fit into one second is $1 \div 0.002 = \mathbf{500 \text{ action potentials per second}}$.
- Repeat it for an absolute refractory period of 1 ms. $1 \div 0.001 = \mathbf{1,000 \text{ per second}}$. A shorter refractory period gives a higher ceiling.
- Read the answer as a sentence: however strong the stimulus is, this axon cannot send more than 500 action potentials in one second, because for 2 ms after each one its sodium channels cannot open.
- This is a ceiling and not a typical rate. Most neurons fire far below it, commonly well under 100 per second, because the relative refractory period comes next and a stimulus has to be unusually strong to fire the axon during it.

Absolute refractory period

About 1 to 2 ms from threshold. Na^+ channels are open and then inactivated, so no stimulus of any size can fire a second action potential.

Relative refractory period

The next few milliseconds. Some Na^+ channels have reset, K^+ channels are still open and the membrane is hyperpolarized. A larger than normal stimulus can fire a smaller than normal action potential.

Inactivated

The state of the voltage gated Na^+ channel after it has opened. It cannot reopen until the membrane repolarizes and the channel resets.

Maximum firing frequency

$1 \div \text{the absolute refractory period}$. With 2 ms it is 500 per second. Most neurons fire far below their ceiling.

How a neuron codes the strength of a stimulus

SLIDES 7 TO 8 OF THE DECK

Because every action potential is the same size, strength cannot be coded in the size of the signal. It is coded in the number of action potentials per second. A stronger stimulus opens more channels in the receptor ending, so it produces a larger graded potential, and one that lasts longer. At the trigger zone, a graded potential that is still above threshold when the absolute refractory period ends fires another action potential, and the larger it is, the sooner in the relative refractory period it can do that. So a larger, longer graded potential produces more action potentials per second. This is frequency coding. The ceiling is the one set by the

absolute refractory period, and the length of the burst codes how long the stimulus lasted.

At the axon terminal, each action potential releases about the same amount of neurotransmitter, so more action potentials per second release more neurotransmitter. That is how the next cell receives the information about strength. The next lecture covers how this release works.

A single axon has only that one way to report a stronger stimulus. The nervous system as a whole has a second way. A stronger stimulus reaches more receptors, including ones with a higher threshold, so more neurons fire. This is called recruitment, or population coding. The brain reads both: how fast each axon is firing, and how many axons are firing. Recruitment matters most at high intensities. An axon with a 2 ms absolute refractory period that is already firing at 400 per second cannot double its rate, because its ceiling is 500, but a stimulus twice as strong still recruits receptors that were silent before.

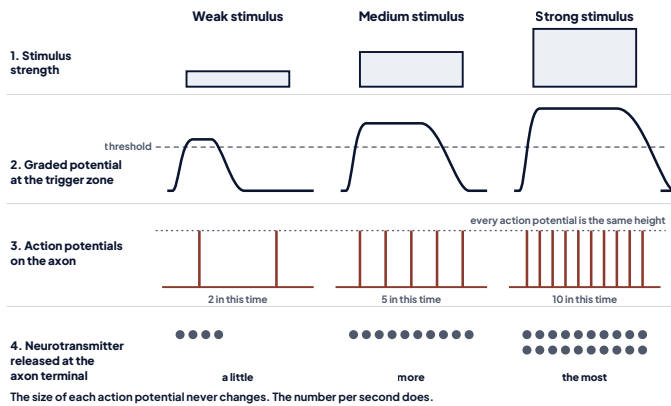


Figure 2. One sensory neuron and three strengths of stimulus. Read each column from top to bottom.

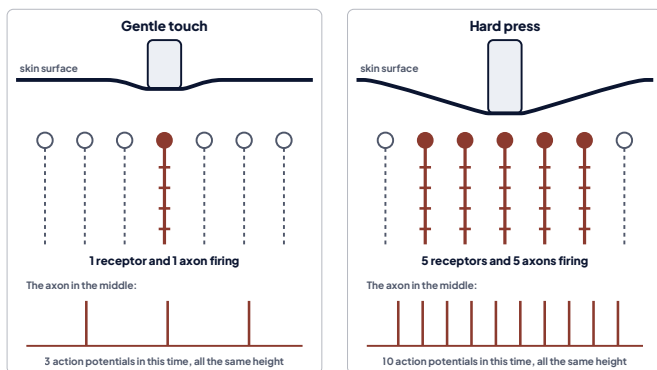


Figure 3. The same patch of skin touched gently and pressed hard.

Frequency coding

Signaling the strength of a stimulus by the number of action potentials per second, because their size is fixed.

Recruitment (population coding)

Signaling a stronger stimulus by the number of neurons that fire.

Trigger zone

The part of the neuron with the highest density of voltage gated Na^+ channels, where graded potentials are turned into action potentials.

How the action potential moves along the axon

SLIDES 9 TO 10 OF THE DECK

Stop a traveling action potential at one moment and look at the membrane. In the active zone, Na^+ has entered and the inside of the membrane is positive, while the neighboring stretches are still negative inside. Positive charge flows along the inside of the axon toward the negative regions on both sides. This flow of charge is called local current. Ahead of the action potential, the local current depolarizes resting membrane. When that patch reaches threshold, its own Na^+ channels open and it fires a full action potential, and then it does the same for the patch after it.

Behind the action potential, the local current arrives at membrane that fired a moment ago. Its Na^+ channels are inactivated and its K^+ channels are open, so it is refractory and cannot fire. By the time it has recovered, the active zone has moved too far away to reach it. That is why an action potential that starts at the trigger zone travels only toward the axon terminal. The axon itself has no preferred direction. If an axon is stimulated in the middle in the laboratory, two action potentials travel away from the electrode in opposite directions, because the membrane on both sides is at rest.

The action potential that reaches the axon terminal is therefore not the one that left the trigger zone. It is a copy, made fresh by each patch of membrane along the way, using the Na^+ and K^+ gradients at that patch. That is what regenerated means, and it is why the signal is the same size a meter away as it was at the start. A graded potential is different. It spreads by local current only, nothing renews it, charge leaks out through the membrane as it goes, and it is gone within a few millimeters. A row of falling dominoes is a fair comparison on two points: each domino falls with its own stored energy, so the last falls as hard as the first, and a domino that has fallen cannot fall again until it is set back up.

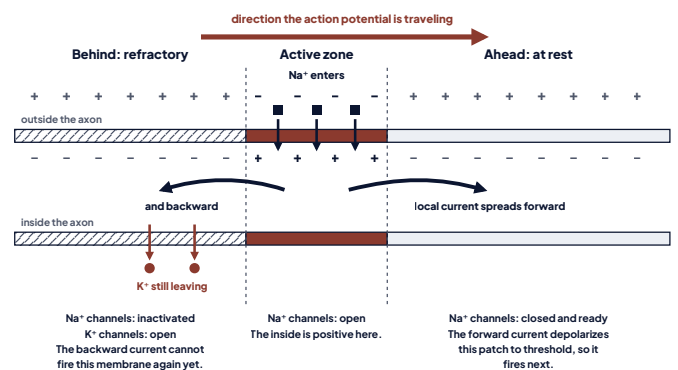


Figure 4. Local current spreads forward and backward. Only the forward direction finds membrane that can fire.

Local current

The flow of positive charge along the inside of the axon from the active zone toward neighboring membrane that is still negative inside.

Regenerated

Made again at each patch of membrane. The action potential does not decrease with distance because each patch fires its own.

Conduction

The movement of an action potential along an axon at full size.

What sets conduction velocity

SLIDES 11 TO 14 OF THE DECK

Two things decide how fast an axon conducts. The first is diameter. Local current flows through the cytoplasm, and a wider axon has more room for charge to flow, so its internal resistance is lower, in the same way that a wide pipe carries water more easily than a narrow one. The local current reaches farther ahead before it fades, so membrane farther down the axon is brought to threshold sooner. The second is how much current leaks out through the membrane on the way, because current that leaks out is lost to the job of depolarizing the next patch. Myelin, which is many layers of glial cell membrane wrapped around the axon, raises the resistance of the membrane so very little leaks out. It also means less charge is needed to change the voltage across that stretch. Both effects let local current spread farther and faster.

Myelin gives far more speed for the space it takes up. The squid giant axon has no myelin. It is about 500 μm across and conducts at about 25 m/s. Your largest myelinated axons are about 20 μm across and conduct at up to about 120 m/s. Area goes with the square of the diameter, and $(500 \div 20)^2$ is 625, so about 625 of your axons fit in the space of one squid axon, and each is nearly five times faster. That is what lets a nerve as thick as a finger carry many thousands of fast axons down a limb.

In an unmyelinated axon, voltage gated channels are spread along the whole membrane, and every patch has to depolarize and fire in turn. This is continuous conduction, and it is slow because opening channels takes time at every patch. In a myelinated axon the voltage gated Na^+ channels are concentrated at the nodes of Ranvier. Local current from one node flows inside the axon, under the myelin, and reaches the next node still strong enough to bring it to threshold, so the action potential is regenerated only at the nodes. This is saltatory conduction, from the Latin word for jumping. Nothing actually jumps. What travels between the nodes is local current, a very fast spread of charge along the inside of the axon. The time is spent at the nodes, and there are far fewer of those stops. It also costs less ATP, because Na^+ enters and K^+ leaves only at the nodes, so the Na^+/K^+ pump has far fewer ions to move back.

Real human nerve fibers conduct at anywhere from about 0.5 to about 120 m/s. Large myelinated A alpha fibers, 12 to 20 μm , conduct at 70 to 120 m/s and carry commands to skeletal muscle. A beta fibers, 5 to 12 μm , carry touch and pressure at 30 to 70 m/s. Thin myelinated A delta fibers, 2 to 5 μm , carry fast, sharp pain and cold at 12 to 30 m/s. Unmyelinated C fibers, 0.4 to 1.2 μm , carry slow, aching pain and warmth at 0.5 to 2 m/s. These are approximate ranges, and books differ a little on the boundaries. Two rough rules reproduce them: for a myelinated axon, velocity in m/s is about 6 times the diameter in μm , and for an unmyelinated axon it is about the square root of the diameter in μm . To rank any set of axons, put the myelinated ones first, and within each group put the larger diameter first.

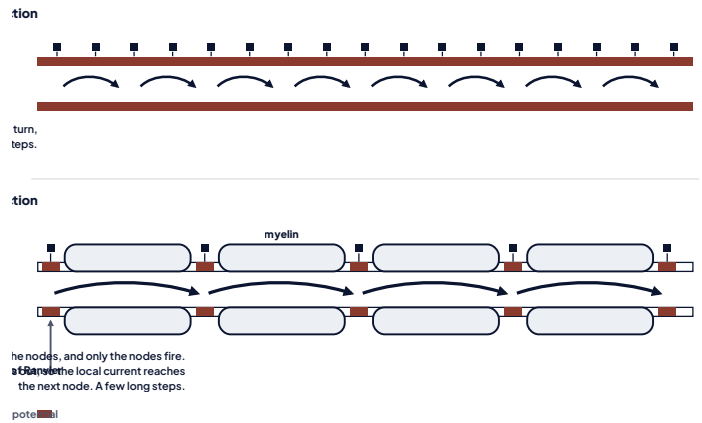


Figure 5. Two axons of the same diameter. The shaded membrane is the membrane that depolarizes and fires.

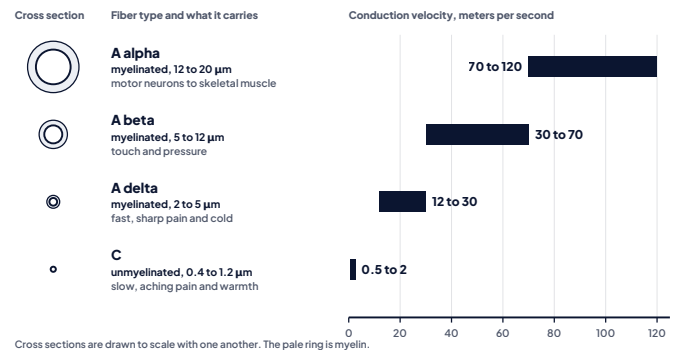


Figure 6. Approximate ranges for human nerve fibers.

WORKED PROBLEM 2

Sensory axons from your toe run about 1 m to the spinal cord. How long does an action potential take in a large myelinated fiber at 100 m/s, and in a small unmyelinated fiber at 1 m/s? Then explain why a stubbed toe hurts twice.

time = distance $\div$ velocity.

THE WORK

- The relationship is time = distance $\div$ velocity.
- Large myelinated fiber: $1 \text{ m} \div 100 \text{ m/s} = 0.010 \text{ s} = \mathbf{10 \text{ ms}}$.
- Small unmyelinated fiber: $1 \text{ m} \div 1 \text{ m/s} = \mathbf{1 \text{ s}}$. That is 100 times longer, for the same stretch of leg.
- Now the two kinds of pain fiber. A delta fibers are thin and myelinated. At about 20 m/s, 1 m takes $1 \div 20 = 0.05 \text{ s} = \mathbf{50 \text{ ms}}$. C fibers are unmyelinated. At about 1 m/s, 1 m takes $\mathbf{1 \text{ s}}$.
- So when you stub your toe, the action potentials in the A delta fibers reach the spinal cord about a second before the ones in the C fibers. You feel a sharp, well located pain first, and a dull, aching, spreading pain about a second later. These are called first pain and second pain. Both sets of action potentials started at the same moment.
- The withdrawal reflex uses the faster fibers, which is why your foot has already moved before the ache begins.

Continuous conduction

Conduction in an unmyelinated axon, where every patch of membrane fires in turn.

Saltatory conduction

Conduction in a myelinated axon, where the action potential is regenerated only at the nodes of Ranvier and local current carries the depolarization between them.

Node of Ranvier

A short bare gap between segments of myelin, where voltage gated Na^+ channels are concentrated.

First pain and second pain

The sharp pain carried by A delta fibers, followed about a second later by the aching pain carried by slower C fibers.

Changing the conditions: myelin, potassium and calcium

SLIDES 16 TO 18 OF THE DECK

Remove the myelin from part of an axon, as happens in the central nervous system in multiple sclerosis. Where myelin is lost, local current leaks out through the bare membrane, so less of it reaches the next node. That node takes longer to reach threshold, and conduction slows. If too little arrives, the node never reaches threshold and conduction fails. The bare stretch cannot make up for it, because it was under myelin and has very few voltage gated Na^+ channels of its own. Warming makes this worse. If body temperature rises from 37.0°C (98.6°F) to 37.5°C (99.5°F), ion channels gate faster, the Na^+ channels at each node inactivate sooner, and each action potential is briefer and sends less current on. A healthy node receives several times more current than it needs, so it is unaffected. A node beyond a demyelinated stretch is receiving barely enough, so it fails. Cooling brings it back, and no new damage has occurred.

Change the potassium in the blood. Plasma K^+ is the outside K^+ for every neuron and muscle cell. In the resting potential lecture you calculated that 7 mM moves the resting potential to about -64 mV and 2.5 mM moves it to about -76 mV . Threshold is set by the Na^+ channels and stays at about -55 mV , so what changes is the gap between rest and threshold. Worked problem 3 follows both cases through.

Change the calcium in the blood. Ca^{2+} ions in the extracellular fluid bind to the outside of the voltage gated Na^+ channel, and while they are bound the channel needs a larger depolarization to open. With low extracellular Ca^{2+} the channels open more easily, so threshold moves closer to the resting potential. Nerves begin to fire without being stimulated, and the muscles they supply contract. The result is tingling, cramps and sustained spasm, called tetany. In Chvostek's sign, tapping the facial nerve in front of the ear makes the face twitch. In Trousseau's sign, a blood pressure cuff kept inflated on the arm for about three minutes brings on a spasm of the hand and wrist. With high extracellular Ca^{2+} the reverse happens, and nerve and muscle are less excitable. Students often expect low calcium to weaken muscle. The Ca^{2+} that makes skeletal muscle contract is stored inside the muscle cell, and a low plasma Ca^{2+} hardly changes that store. It also explains why calcium given into a vein helps in hyperkalemia: it moves threshold away from the depolarized resting potential and restores the gap.

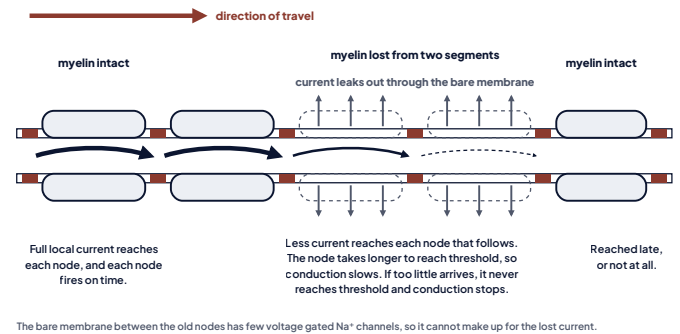


Figure 7. Demyelination. Less current reaches each node beyond the bare stretch.

Plasma potassium moves the resting potential. Threshold stays at about -55 mV .

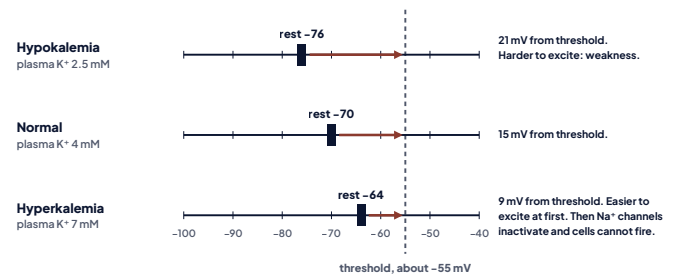


Figure 8. The gap between the resting potential and threshold at three plasma potassium values.

WORKED PROBLEM 3

For a plasma K^+ of 2.5 mM, 4 mM and 7 mM, work out how far the resting potential is from threshold and predict what happens to excitability. Explain both stages of hyperkalemia.

Resting potentials from the resting potential lecture: about -76, -70 and -64 mV. Threshold about -55 mV.

THE WORK

- 1 Normal, 4 mM: the resting potential is about -70 mV and threshold is about -55 mV, so the gap is **15 mV**. A graded potential has to depolarize the trigger zone by 15 mV to fire the neuron.
- 2 Hypokalemia, 2.5 mM: from the resting potential lecture, the resting potential is about -76 mV. The gap to threshold is $-55 - (-76) = 21 \text{ mV}$. The same graded potential that used to reach threshold now falls short, so neurons and skeletal muscle are **less excitable**. The patient is weak.
- 3 Hyperkalemia, 7 mM: the resting potential is about -64 mV. The gap is $-55 - (-64) = 9 \text{ mV}$. At first a smaller graded potential is enough, so cells are **more excitable**.
- 4 Now add what you know about the sodium channel. A membrane that stays depolarized holds some of its voltage gated Na^+ channels in the inactivated state, and an inactivated channel cannot open. The longer and the further the membrane is depolarized, the fewer channels are closed and ready.
- 5 With too few channels ready, the action potential rises slowly and is small, and if enough are inactivated the cell cannot fire at all. So hyperkalemia makes cells more excitable first and then less excitable, and what changes in between is the number of Na^+ channels that are available.
- 6 The Goldman equation gives -71 mV for the normal resting potential with the values from that lecture. It is rounded to -70 mV here. Books also vary by a few millivolts on threshold.

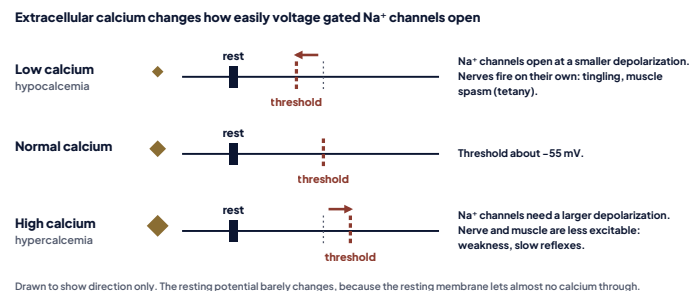


Figure 9. Extracellular calcium changes how easily the Na^+ channels open. Direction only.

Demyelination

Loss of myelin from part of an axon. Current leaks out between nodes, and conduction slows or fails.

Hyperkalemia

High plasma K^+ . The resting potential depolarizes toward threshold. Cells are more excitable at first, and then Na^+ channels inactivate and cells cannot fire.

Hypokalemia

Low plasma K^+ . The resting potential hyperpolarizes, farther from threshold, and cells are less excitable.

Tetany

Sustained muscle spasm caused by nerves firing on their own, as in low extracellular Ca^{2+} .

Blocking the sodium channel, and the dentist's chair explained

SLIDES 19 TO 21 OF THE DECK

There are two different ways to interfere with Na^+ entry, and they give different results. Lower the Na^+ outside the axon, and E_{Na} falls, so the peak of the action potential is lower, but the channels still open and the action potential is still conducted. Block the channel, and there is no action potential at all. Local anesthetics such as lidocaine cross the membrane, enter the voltage gated Na^+ channel from the inside of the cell and block the pore. Tetrodotoxin blocks the same channel from the outside.

Now the dentist's chair. The pain receptors and the first stretch of axon work normally, so graded potentials still form and action potentials still start and travel toward the brainstem. In the injected stretch the Na^+ channels cannot open. Lidocaine binds most readily to channels that are open or inactivated, so axons that are firing are blocked sooner. Local current from the last working patch still flows into the blocked stretch, but no channels open, so it is only a graded depolarization and it fades within a few millimeters. Nothing reaches the brainstem, so there is no pain.

Small fibers are blocked before large ones. To stop a myelinated axon, the drug has to block about three nodes in a row. In thin A delta fibers the nodes are close together, so a short stretch of drug covers enough of them. In thick A beta fibers, which carry pressure, the nodes are up to a millimeter or two apart, and the same stretch of drug may not cover three. C fibers are thin and have no myelin to keep the drug out. That is why the patient feels pushing and no pain. This order is the usual teaching rule and it is a simplification, because where a fiber lies within the nerve and how often it is firing also matter. Everything returns when the drug diffuses away, because nothing was damaged.

Look at it as an experiment. The problem is that action potentials from the tooth reach the brain. The target variable is the number of Na^+ channels that can open in one stretch of axon. The intervention is a drug that blocks them. The mechanism is that the action potential cannot be regenerated in that stretch. The same mechanism predicts what happens if a large amount of the drug reaches the blood. Every neuron and every heart muscle cell has these channels, so the first effects are in the brain, with numbness around the mouth, ringing in the ears and then seizures, and in the heart conduction slows. This is why a local anesthetic is kept local.

Every perturbation in this lecture acted on the same thing: how many voltage gated Na^+ channels are closed and ready to open. Hyperkalemia inactivates them. Calcium changes how easily they open. A local anesthetic blocks them. Demyelination reduces the current that reaches them.

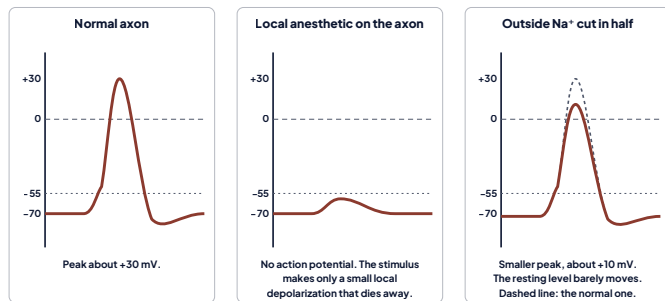


Figure 10. The same axon and the same stimulus under three conditions.

WORKED PROBLEM 4

An axon is bathed in fluid with 72.5 mM Na^+ , half the normal 145 mM. Inside stays at 15 mM. Calculate the new E_{Na} , predict what happens to the peak of the action potential, and compare that with an axon whose Na^+ channels are blocked.

$$E_{\text{ion}} = (61.5 \text{ mV} \div z) \times \log_{10}(\text{outside} \div \text{inside}), \text{ with } z = +1.$$

THE WORK

- Normal sodium: $E_{\text{Na}} = 61.5 \times \log_{10}(145 \div 15) = 61.5 \times 0.985 = +61 \text{ mV}$.
- Half the outside sodium: $E_{\text{Na}} = 61.5 \times \log_{10}(72.5 \div 15) = 61.5 \times 0.684 = +42 \text{ mV}$. Halving the outside concentration lowers E_{Na} by $61.5 \times \log_{10}(2) = 18.5 \text{ mV}$.
- At the peak of an action potential the membrane is far more permeable to Na^+ than to anything else, so the peak follows E_{Na} . When E_{Na} drops by about 19 mV, the peak drops by about the same amount, from about +30 mV to about +10 mV. Hodgkin and Katz did this experiment on the squid giant axon in 1949, and the peak fell by about as much as the Nernst equation predicts.
- The resting potential changes only a little, because the resting membrane is barely permeable to Na^+ . You saw the same thing in the resting potential lecture.
- The action potential is smaller, but it is still there. The channels still open, Na^+ still enters, and each patch of membrane still fires the next. With a local anesthetic or tetrodotoxin the channels cannot open at all, so there is no action potential to conduct.

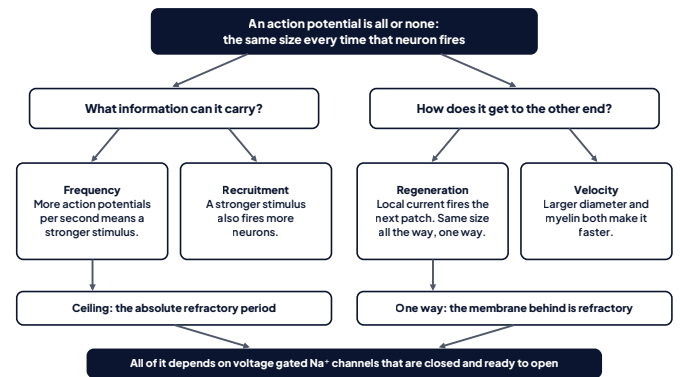


Figure 11. The whole lecture on one map. Read it from the top down.

Local anesthetic

A drug such as lidocaine that blocks voltage gated Na^+ channels from the inside of the cell, so no action potential is conducted past the blocked stretch.

Tetrodotoxin

A toxin that blocks the voltage gated Na^+ channel from the outside.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ and Ca^{2+}
 Opposite charges attract, and current is a flow of charge
 Resistance: the harder it is for charge to flow, the higher the resistance

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential, the Nernst equation, and how plasma K^+ moves the resting potential (this week)
 Neurons and glia: axon, trigger zone, axon terminal, myelin and nodes of Ranvier (this week)
 Graded potentials and the action potential: threshold, all or none, the two gates of the Na^+ channel, tetrodotoxin (this week)

The anatomy you need is from the neurons and glia lecture: axon, trigger zone, axon terminal, myelin and node of Ranvier.

MATH

Dividing by a decimal: $1 \div 0.002 = 500$
 Changing milliseconds to seconds: $2 \text{ ms} = 0.002 \text{ s}$
 $\text{time} = \text{distance} \div \text{velocity}$
 The log button, for worked problem 4

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Competencies 19 and 20, the action potential simulation and the

nerve conduction measurement, are done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

15

Coding of stimulus intensity

Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

Do one of the two. Head your paper with the number.

15-1

Draw three stimulus bars of increasing strength across the bottom of your box, weak, medium, and strong. Above each, draw the spike train one sensory neuron sends, as vertical marks on a time line. Keep every spike exactly the same height. One line: say precisely what changed between your three trains, and what could not have changed.

15-2

Draw a patch of skin with a stimulus pressed into it, gentle on the left and hard on the right. In each, draw how many receptors and how many axons get recruited, and draw the spike train on one of those axons underneath. One line: the nervous system has two ways to say louder, so name both from your drawing and say which one a single axon can do on its own.

16

Refractory periods

Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

Do one of the two. Head your paper with the number.

16-1

Draw an action potential tracing and bracket the absolute and relative refractory periods underneath it. Along each bracket, draw the state of the voltage gated sodium channels, inactivated and then recovering, and show whether the potassium channels are still open. Then draw a second stimulus arriving inside each period and show the result. One line: say why no stimulus at all, however large, works during the absolute period.

16-2

Draw an axon with an action potential halfway along it and an arrow for the direction it is traveling. Shade the membrane behind it and mark what state its sodium channels are in. Draw the local current spreading both forward and backward, and show why only one direction succeeds. One line: an absolute refractory period of about 1 to 2 ms sets a ceiling on firing rate, so work out that ceiling in spikes per second.

17

Conduction velocity

Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

Do one of the two. Head your paper with the number.

17-1

Draw four axons stacked vertically: large myelinated, small myelinated, large unmyelinated, and small unmyelinated. Rank them fastest to slowest and write a rough velocity beside each, using the real range of about 0.5 up to about 120 m per second. One line: say which buys more speed per unit of space taken up, diameter or myelin, and why that matters for a nerve running down a limb.

17-2

Draw one myelinated and one unmyelinated axon of similar diameter side by side, the same length. On each, shade where the membrane actually depolarizes, continuously along one and node to node on the other. Draw the local current loops on both. One line: saltatory means jumping, but the action potential does not really jump, so say what is actually moving between the nodes.

18

Effects of altered extracellular ions

Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

Do one of the two. Head your paper with the number.

18-1

Draw a voltage scale from minus 100 to plus 30 mV with the resting potential at minus 70 and threshold at about minus 55 marked on it. Using the Nernst logic for potassium, draw where the resting potential moves in hyperkalemia and in hypokalemia, and show the remaining gap to threshold in each. One line: hyperkalemia first makes the cell more excitable and then less, so say what happens to the sodium channels in between.

18-2

Draw a normal action potential tracing, then draw the same axon after a local anesthetic such as lidocaine has blocked its voltage gated sodium channels, and show what is left of the trace. Beside it, draw the axon in low extracellular sodium and show what changes about the peak, not about the resting level. One line: say why blocking the channel abolishes the signal while low extracellular sodium only shrinks it.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · How action potentials carry information. Velocities, diameters and times are approximate teaching values. All figures are original drawings for this course.

The chemical synapse

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 21 TO 23 OF WEEK 4

20 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

An action potential reaches the end of an axon and stops there. The next cell is across a gap of extracellular fluid, and an electrical signal cannot cross that gap. In this lecture we follow how the signal is turned into a chemical, carried across, and turned back into an electrical signal on the other side. The place where this happens is a chemical synapse. We take the events in strict order, because each step causes the next one, and once you know the order you can predict what happens when any single step fails.

Here is the problem we solve by the end. A five month old girl has been constipated for several days. Now she feeds poorly, her cry is weak, her eyelids droop, and she cannot hold up her head. Within two days she needs a ventilator to breathe. She is alert and she feels every touch. A week earlier her pacifier had been dipped in honey.

Testing narrows it down. She is awake, follows faces and grimaces when her foot is pinched, so her sensory neurons and her brain are working. Action potentials travel down her motor axons at normal speed, so conduction is working. But when a motor nerve is stimulated, the electrical response of the muscle is far smaller than normal. In laboratory studies of muscle poisoned in this way, acetylcholine applied straight onto the muscle still makes it contract normally, so the muscle and its receptors work. Her gut, which is also driven by neurons that release acetylcholine, is hardly moving.

Put those results together. The axon conducts, and the muscle responds when acetylcholine reaches it. So the failure lies between the action potential arriving at the end of the axon and acetylcholine arriving at the muscle's receptors. That stretch is the synapse, and this lecture divides it into nine steps. As you read, decide which step has failed in this baby.

See it move: run one synapse yourself

SLIDE 15 OF THE DECK

This is a working model of the synapse in this lecture. Press Send one action potential and follow the step that is named under the picture. The table on the right reports what happened in numbers, and the graph shows the response of the postsynaptic cell. Then block one step at a time. Each block matches a toxin or drug described later in these notes, so keep the model open and come back to it as you read. The note under the model says which parts are real and which are illustrative.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-synapse-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-synapse-notes.html)

THREE THINGS TO TRY

- 1 Send one action potential with nothing blocked and read the table. Then turn on Botulinum toxin and send another. Calcium still enters, but vesicles released falls from 3 to 0 and the response falls from 1.00 to 0. Now turn that off and block the Ca^{2+} channels. The postsynaptic cell is just as silent, but this time calcium entry reads no. The readouts upstream of the block are what tell you where the sequence stopped.
- 2 Clear the blocks and block the removal enzyme. The peak of the response rises from 1.00 to about 1.80 and it lasts about 39 ms where it lasted about 28 ms before. Block reuptake as well, so that only diffusion is left, and it rises to about 3.69 and lasts about 75 ms. Compare the solid line on the graph with the dashed line each time.
- 3 Clear the blocks, send a steady train, and move the slider from 10 to 100 action potentials per second. The average response over the train rises from about 0.14 to about 1.35, because more action potentials per second release more transmitter per second. Then block the removal enzyme at 100 per second and watch the response stop returning to baseline.

The parts of a chemical synapse

SLIDE 3 OF THE DECK

A synapse is the place where a neuron passes its signal to another cell. The cell sending the signal is the presynaptic cell, and the cell receiving it is the postsynaptic cell, which may be another neuron, a muscle cell or a gland cell. A chemical synapse has three parts.

The first part is the axon terminal of the presynaptic neuron. This is the output zone from the lecture on neurons. It holds synaptic vesicles, which are small membrane sacs filled with neurotransmitter. Some vesicles are docked against the

membrane that faces the next cell, ready to release, and others wait in reserve farther back. The terminal has mitochondria to supply ATP, and its membrane contains voltage gated Ca^{2+} channels, which are closed at rest.

The second part is the synaptic cleft, a gap of extracellular fluid about 20 to 40 nm wide between neurons, and a little wider at the neuromuscular junction. The two membranes do not touch, which is why the action potential cannot continue across.

The third part is the postsynaptic membrane, which holds receptors. A receptor is a protein that binds one particular transmitter. Astrocytes wrap many synapses in the central nervous system, and they take up some transmitters from the cleft, as you saw in the lecture on glia.

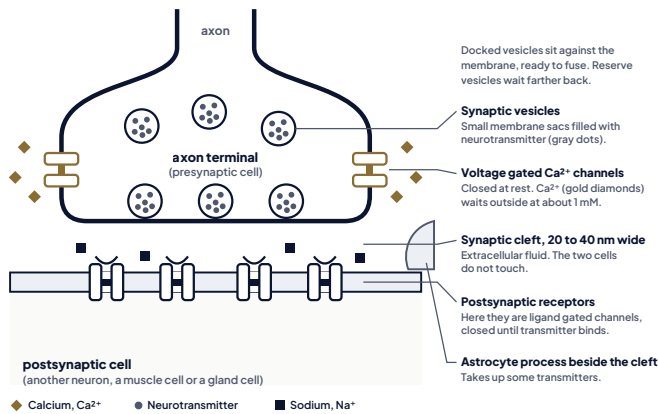


Figure 1. The presynaptic terminal, the synaptic cleft and the postsynaptic membrane. The cleft is drawn much wider than it really is.

Synapse

The place where a neuron passes its signal to another cell.

Presynaptic and postsynaptic

The presynaptic cell sends the signal. The postsynaptic cell receives it.

Synaptic vesicle

A small membrane sac in the axon terminal, filled with neurotransmitter.

Synaptic cleft

The gap of extracellular fluid between the two cells, about 20 to 40 nm wide.

Receptor

A protein in the postsynaptic membrane that binds one particular transmitter and starts the response.

Steps 1 to 5: inside the axon terminal

SLIDES 4 TO 5 OF THE DECK

Step 1. An action potential travels down the axon and depolarizes the membrane of the terminal. Step 2. That depolarization opens voltage-gated Ca^{2+} channels. These channels respond to voltage in the same way as the voltage-gated sodium channels of the axon, but the ion they pass is calcium.

Step 3. Ca^{2+} enters the terminal. Free calcium is about 1 mM outside the cell and about 0.0001 mM inside, which is a 10,000 fold difference, and the inside of the cell is negative. Both the concentration gradient and the voltage drive calcium in. Worked problem 1 puts numbers on this.

Step 4. The Ca^{2+} that enters binds a sensor protein called synaptotagmin on vesicles that are already docked at the membrane. Step 5. That triggers the SNARE proteins. One SNARE protein sits on the vesicle, and its partners sit on the terminal membrane. They wind around each other and pull the two membranes together until they fuse. The inside of the vesicle is now open to the cleft, and the transmitter empties out. Release by fusion of a vesicle with the cell membrane is exocytosis, which was covered in Week 2. Afterward the vesicle membrane is taken back in by endocytosis and refilled.

Calcium is the link between the electrical signal and the chemical one. Without Ca^{2+} entry there is no release, however many action potentials arrive. It also explains how the frequency code from the last lecture is passed on: more action potentials per second let in more Ca^{2+} , more vesicles fuse, and more transmitter is released.

One classic experiment shows that calcium entry is required. Put the terminal in a bath with no calcium and send an action potential. Steps 1 and 2 happen normally, because the channels respond to voltage and not to calcium. Step 3 fails, because there is no Ca^{2+} outside to enter, and nothing after it happens.

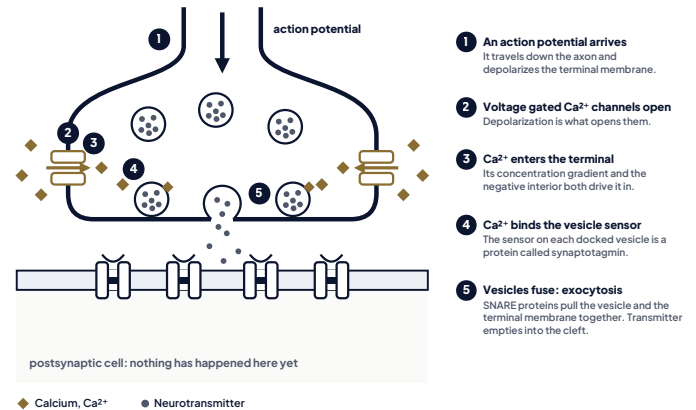


Figure 2. The first five steps. Each one causes the next.

WORKED PROBLEM 1

Free Ca^{2+} is about 1 mM outside the terminal and about 0.0001 mM inside. Find E_{Ca} . Then find the driving force on Ca^{2+} at rest, -70 mV, and at the peak of an action potential, +30 mV, and say which way Ca^{2+} is driven each time.

Use the Nernst equation from the resting potential lecture.
Driving force = membrane potential – equilibrium potential of the ion. For a positive ion, a negative answer means the ion is driven into the cell.

THE WORK

- 1 Write the ratio, outside over inside. $1 \div 0.0001 = 10,000$. The log of 10,000 is 4.
- 2 Calcium carries two positive charges, so $z = +2$. $E_{\text{Ca}} = (61.5 \div 2) \times 4 = 30.75 \times 4 = +123 \text{ mV}$. This is the same value as on the voltage scale in the resting potential lecture.
- 3 The driving force on an ion is the membrane potential minus that ion's equilibrium potential. At rest: $-70 - (+123) = -193 \text{ mV}$. For a positive ion, a negative driving force means the ion is driven **into** the cell.
- 4 At the peak of the action potential: $+30 - (+123) = -93 \text{ mV}$. That is smaller, but it still points inward. The membrane never gets anywhere near +123 mV, so at every moment of an action potential calcium is driven inward.
- 5 So why does calcium not enter at rest, when the driving force is largest? Because its channels are closed. The gradient is always there. The action potential supplies the one thing that was missing, which is an open channel.
- 6 Because there is so little free calcium inside to begin with, a small amount entering raises the concentration right beside the channels a hundredfold or more within a fraction of a millisecond. A signal that starts from almost nothing and rises that fast is easy for the cell to detect, and the vesicle sensor sits right beside those channels.

Voltage gated Ca^{2+} channel

A channel in the terminal membrane that opens when the membrane depolarizes and lets calcium in.

Synaptotagmin

The calcium sensor on a docked vesicle. When calcium binds it, fusion is triggered.

SNARE proteins

Proteins on the vesicle and on the terminal membrane that pull the two membranes together so that they fuse.

Exocytosis

Release of a vesicle's contents by fusion of the vesicle with the cell membrane.

Driving force

Membrane potential minus an ion's equilibrium potential. It tells you how hard, and which way, the ion is driven when a channel opens.

Steps 6 to 9, the delay, and the direction

SLIDES 6 TO 7 OF THE DECK

Step 6. The transmitter diffuses across the cleft. Nothing carries it. Because the gap is only 20 to 40 nm, diffusion takes about a microsecond. Step 7. The transmitter binds receptors on the postsynaptic membrane. Each receptor fits one transmitter, the way a lock fits one key, and the binding is reversible: a transmitter molecule binds, stays for a moment, and comes off again.

Step 8. The postsynaptic cell responds. In the simplest case the receptor is a ligand gated channel, which was covered in the resting potential lecture. With transmitter bound, it opens, Na^+ enters, and the membrane depolarizes a little. That response is a graded potential. Its size depends on how much transmitter was released, it fades with distance, and it has no threshold of its own. Whether it leads to an action potential in the postsynaptic cell is the subject of the next lecture.

Step 9. The transmitter is removed from the cleft. As the concentration in the cleft falls, the receptors empty, the channels close, and the signal ends. The three routes of removal have their own section below.

All of this takes time. The postsynaptic response begins about 0.5 to 1 ms after the action potential reaches the terminal. This is the synaptic delay. Most people guess that the delay is the time taken to diffuse across the cleft, but that is only about a microsecond, less than one percent of it. Most of the delay is spent in the terminal: the voltage gated Ca^{2+} channels take time to open, and then Ca^{2+} has to trigger fusion.

A chemical synapse also works in one direction only. Only the presynaptic side has vesicles ready to release, and only the postsynaptic side has the receptors facing them.

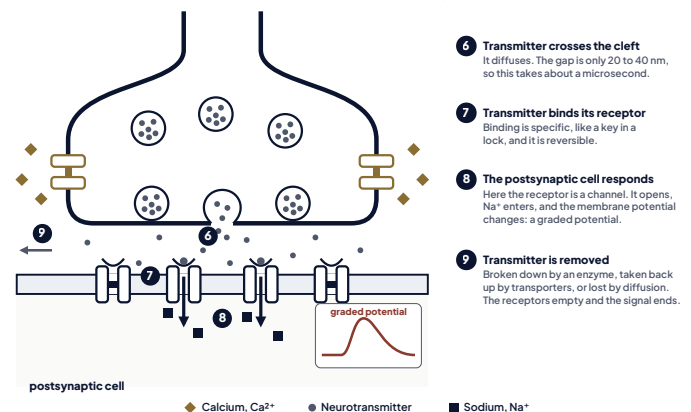


Figure 3. The last four steps. The response in the postsynaptic cell is a graded potential.

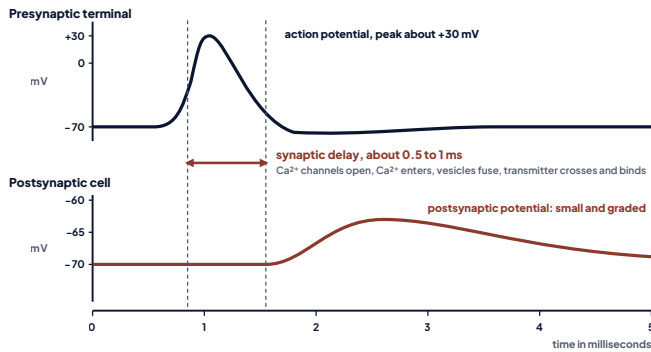


Figure 4. The action potential in the terminal and the response in the postsynaptic cell, on one time axis. Voltages are typical teaching values.

WORKED PROBLEM 2

A pathway has 1 m of axon in total, conducting at 50 m/s. How long does the signal take if the pathway contains 1 synapse? How long if it contains 5?

Use 1 ms for each synaptic delay.

THE WORK

- 1 Conduction time is distance divided by speed. $1 \text{ m} \div 50 \text{ m/s} = 0.02 \text{ s} = \mathbf{20 \text{ ms}}$.
- 2 One synapse adds about 1 ms. Total: $20 + 1 = \mathbf{21 \text{ ms}}$.
- 3 Five synapses add about 5 ms. Total: $20 + 5 = \mathbf{25 \text{ ms}}$. The same length of axon now takes about one fifth longer.
- 4 Put the delay in terms of distance. In 1 ms this axon conducts $50 \text{ m/s} \times 0.001 \text{ s} = 0.05 \text{ m}$. Each synapse costs as much time as **5 cm** of axon, although the cleft itself is only 20 to 40 nm across.
- 5 This is why the fastest reflexes use the fewest synapses. The knee jerk reflex has a single synapse in the spinal cord between the sensory neuron and the motor neuron.
- 6 The delay is a cost of using a chemical synapse. What the cell gets in return is a place where the signal can be made larger or smaller, which is the subject of the next lecture.

Graded potential

A change in membrane potential whose size depends on the strength of the stimulus, here the amount of transmitter. It fades with distance.

Synaptic delay

The time between the action potential reaching the terminal and the start of the postsynaptic response, about 0.5 to 1 ms.

and is over in milliseconds. The nicotinic acetylcholine receptor, the AMPA receptor for glutamate and the GABA-A receptor are examples.

A metabotropic receptor is not a channel. With transmitter bound, it activates a G protein on the inner face of the membrane, which is why it is also called a G protein coupled receptor. The G protein may act on a nearby ion channel directly, or it may switch on an enzyme that makes a second messenger such as cAMP. Second messengers can open channels, close channels, or change other proteins in the cell. More steps make the response slower to start, and it can last from hundreds of milliseconds to minutes. One receptor activates many G proteins, so the signal is also amplified. The muscarinic acetylcholine receptor, every receptor for norepinephrine and dopamine, and most receptors for serotonin are metabotropic.

Now the most important idea in this section. The receptor, and not the transmitter, decides the response. Acetylcholine shows this clearly. At skeletal muscle it binds the nicotinic receptor, whose channel passes both Na^+ and K^+ . A muscle fiber rests near -90 mV , so the driving force on Na^+ is $-90 - 61 = -151 \text{ mV}$, inward, and the driving force on K^+ is $-90 - (-95) = +5 \text{ mV}$, outward. Sodium is driven about 30 times harder, so much more Na^+ enters than K^+ leaves, and the muscle fiber depolarizes and contracts.

At the pacemaker cells of the heart, the same acetylcholine binds the muscarinic receptor. Its G protein opens K^+ channels, K^+ leaves, the cell moves away from threshold, and the pacemaker fires less often, so the heart rate slows. That is one of the actions of the muscarinic receptor in the heart and the easiest one to draw. We return to the others in the cardiac weeks.

So the transmitter is only the message that something has arrived. What the postsynaptic cell does about it depends on which receptor it has put in its membrane and on which ions move as a result. If the result brings the cell toward threshold, the synapse is excitatory. If it holds the cell away from threshold, the synapse is inhibitory. It is like a whistle at a sports field: the same whistle tells one team to start and the other to stop, and the meaning is in the listener.

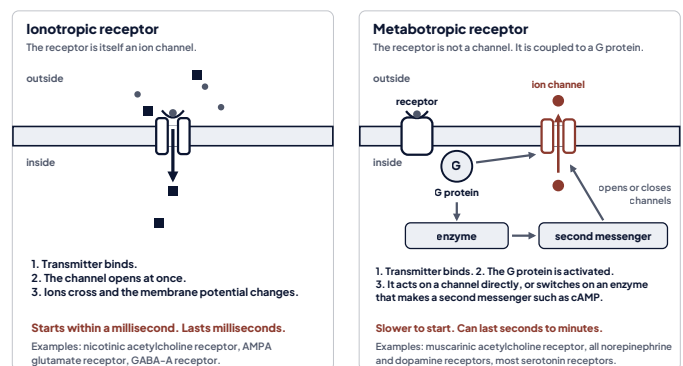


Figure 5. An ionotropic receptor is a channel. A metabotropic receptor works through a G protein.

Two families of receptor, and why the receptor decides

SLIDES 8 TO 9 OF THE DECK

Receptors for neurotransmitters come in two families. An ionotropic receptor is a ligand gated ion channel: the receptor and the channel are the same protein. Transmitter binds, the pore opens, and ions cross. The response begins within a millisecond

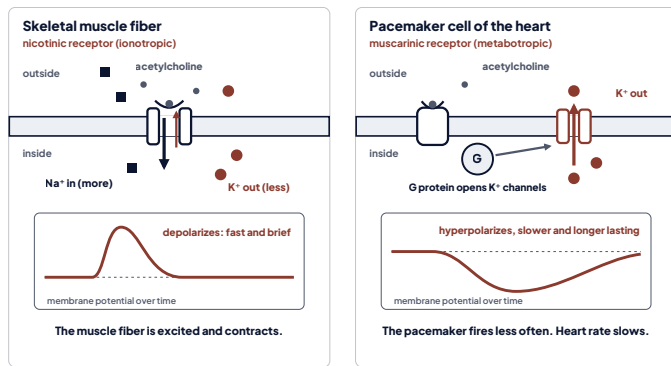


Figure 6. Acetylcholine arriving at two different cells.

Ionotropic receptor

A receptor that is itself an ion channel. Fast and brief.

Metabotropic receptor

A receptor coupled to a G protein. It may act through a second messenger. Slower to start and longer lasting.

Second messenger

A small molecule made inside the cell, such as cAMP, that carries the signal from the receptor to its targets.

Nicotinic and muscarinic receptors

The two families of acetylcholine receptor. Nicotinic receptors are ionotropic. Muscarinic receptors are metabotropic.

Seven classes of neurotransmitter

SLIDES 10 TO 12 OF THE DECK

Silverthorn groups neurotransmitters into seven classes by chemical structure. Acetylcholine is a class of its own. The amines are each made from a single amino acid: dopamine, norepinephrine and epinephrine from tyrosine (these three are the catecholamines), serotonin from tryptophan, and histamine from histidine. The amino acids do most of the fast signaling in the central nervous system: glutamate is the main excitatory transmitter, GABA is the main inhibitory transmitter of the brain, and glycine is the main inhibitory transmitter of the spinal cord. The purines are ATP and adenosine.

Peptides are short chains of amino acids. Substance P excites pain pathways in the spinal cord, and the opioid peptides, the endorphins and enkephalins, inhibit them. Morphine acts on the same receptors. A neuron often releases a peptide together with a small transmitter. The last two classes are unusual. Gases such as nitric oxide, and lipids such as the endocannabinoids, are made at the moment they are needed. They are not stored in vesicles, and they leave the cell by diffusing through its membrane.

The table gives the usual effect of each transmitter and one real place where it acts. Usual effect means the effect at its most common receptors, and the receptor has the final say. The GABA_A receptor and the glycine receptor are Cl⁻ channels. Opening them holds the membrane away from threshold, and the next lecture explains how. A few more facts are worth knowing. Epinephrine is mainly a hormone from the adrenal medulla and a transmitter in only a few brain pathways. Histamine neurons in the hypothalamus promote wakefulness, which is why older antihistamines cause drowsiness. Adenosine is inhibitory in the brain, and caffeine blocks its receptors.

Where a transmitter is made matters. Acetylcholine, the amines, the amino acids and the purines are small molecules. They are made in the axon terminal and loaded into vesicles there. The enzymes that make them are proteins, so those are built in the cell body and shipped down the axon. Peptides are built on ribosomes like any other protein, so they are made in the cell body, packaged into vesicles there, and carried to the terminal by fast axonal transport along microtubules, which was covered in the lecture on neurons. Nitric oxide is made from the amino acid arginine by an enzyme. It lasts only a few seconds, and it acts on whichever cells it diffuses into.

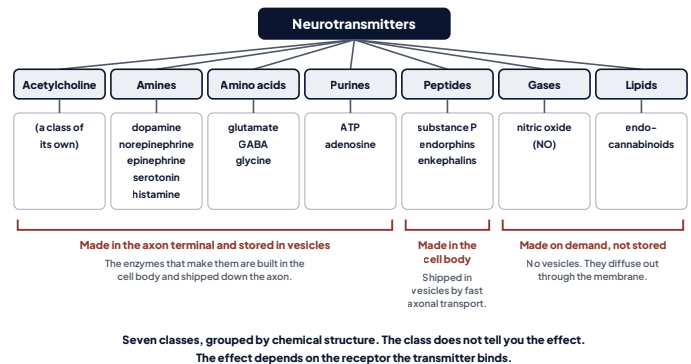


Figure 7. The seven classes, grouped by chemical structure, with where each one is made.

EACH TRANSMITTER, ITS USUAL EFFECT, AND ONE PLACE IT ACTS

TRANSMITTER	USUAL EFFECT	A REAL SITE OF ACTION
Acetylcholine	Excitatory at nicotinic receptors. Excitatory or inhibitory at muscarinic receptors.	Neuromuscular junction (excites skeletal muscle). Heart pacemaker (slows it). Autonomic ganglia and the brain.
Norepinephrine (a catecholamine)	Either one. Every receptor for it is metabotropic.	Released by sympathetic neurons. Speeds the heart through beta-1 receptors and relaxes airway smooth muscle through beta-2 receptors. Arousal pathways in the brain.
Dopamine (a catecholamine)	Either one. Every receptor for it is metabotropic.	Brain: pathways for movement in the basal nuclei, and for reward.
Serotonin	Either one. Most of its receptors are metabotropic.	Brainstem neurons that project widely through the brain: mood, sleep and appetite.
Glutamate	Excitatory. The main excitatory transmitter of the central nervous system.	Brain and spinal cord. Its AMPA and NMDA receptors are cation channels.
GABA	Inhibitory. The main inhibitory transmitter of the brain.	Brain. Its GABA-A receptor is a Cl^- channel, which holds the membrane away from threshold.
Glycine	Inhibitory. The main inhibitory transmitter of the spinal cord.	Spinal cord and brainstem. Its receptor is also a Cl^- channel.
Neuropeptides	Depends on the peptide. All of their receptors are metabotropic.	Substance P excites pain pathways in the spinal cord. Endorphins and enkephalins inhibit them.

WORKED PROBLEM 3

A motor neuron has its cell body in the spinal cord and its terminal in the foot, 1 m away. Fast axonal transport moves vesicles at up to about 400 mm per day. How long does a new supply of a peptide take to arrive? Compare that with the supply of acetylcholine.

Time is distance divided by speed. Keep the units the same.

THE WORK

- Convert the axon length to the units of the transport speed. 1 m = **1,000 mm**.
- Time = distance ÷ speed = 1,000 mm ÷ 400 mm per day = **2.5 days**. That is the fastest case. Any vesicle that moves more slowly takes longer.
- Now acetylcholine. The terminal makes it on the spot from choline and acetyl CoA, and about half of the choline is recovered from the cleft after each release. Emptied vesicle membrane is taken back in by endocytosis and refilled within about a minute.
- So the two supplies are refilled on very different time scales: about a minute for acetylcholine, and days for a peptide.
- Prediction: a neuron that fires rapidly for a long time can run low on its peptide long before it runs low on its small transmitter. A peptide used up at the terminal can only be replaced by the cell body, 2.5 days away.

Catecholamines

Dopamine, norepinephrine and epinephrine. All three are made from the amino acid tyrosine.

Glutamate

The main excitatory transmitter of the central nervous system.

GABA and glycine

The main inhibitory transmitters: GABA in the brain, glycine in the spinal cord.

Neuropeptide

A transmitter that is a short chain of amino acids, made in the cell body and shipped to the terminal.

Ending the signal: three routes of removal

SLIDES 13 TO 14 OF THE DECK

A signal carries information only if it ends. If transmitter stayed in the cleft, the receptors would stay bound, and the postsynaptic cell could not tell one action potential from the next. A doorbell whose button sticks tells you nothing about how many visitors there are. Three routes clear transmitter from the cleft, and each synapse relies mainly on one of them.

The first route is breakdown by an enzyme. Acetylcholinesterase is anchored in the cleft, and it splits acetylcholine into choline and acetate within milliseconds. Neither piece activates the receptor. A transporter carries the choline back into the terminal, where it is joined to acetyl CoA to make new acetylcholine.

The second route is reuptake. Transporter proteins in the presynaptic membrane carry whole transmitter molecules back into the terminal, where they are reloaded into vesicles or broken down. Serotonin, norepinephrine and dopamine are removed mainly this way. Glutamate and GABA are taken up by astrocytes as well as by neurons. These transporters are cotransporters that use the Na^+ gradient, the same kind of secondary active transport covered in Week 2.

The third route is diffusion. Some transmitter always drifts out of the open edges of the cleft. For peptides this is the main route, and peptidases outside the cleft then break them down.

Figure 9 shows what happens to the response to one action potential when removal is blocked, using the model from this lecture. Release is unchanged, so the same amount of transmitter enters the cleft. It then stays longer. Receptors that would have emptied are bound again and again, so the response keeps growing for longer, which makes it taller, and it takes longer to end, which makes it wider. In the model the peak goes from 1.00 to about 1.36 with reuptake blocked and to about 1.80 with the enzyme blocked. Those sizes are illustrative. Which block matters more at a real synapse depends on which route that synapse mainly uses.

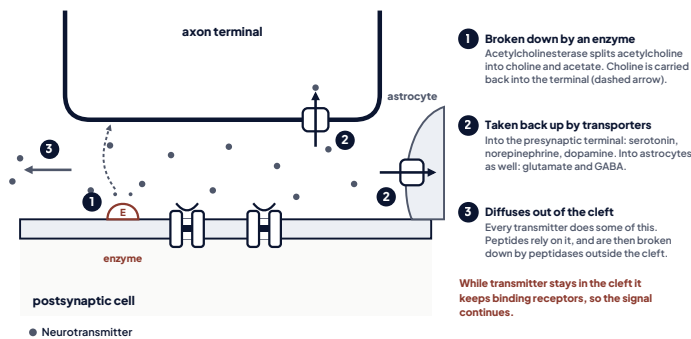


Figure 8. Breakdown by an enzyme, reuptake by transporters, and diffusion out of the cleft.

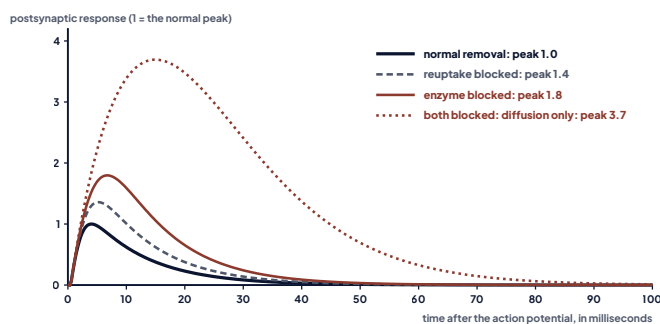


Figure 9. The response to one action potential with each removal route blocked. Relative units from the model. The shapes are illustrative.

Acetylcholinesterase

The enzyme in the cleft that splits acetylcholine into choline and acetate.

Reuptake

Transport of whole transmitter molecules back into the presynaptic terminal, or into glia.

Diffusion

Movement of transmitter out of the cleft down its concentration gradient.

Toxins and drugs as experiments on single steps SLIDES 16 TO 18 OF THE DECK

Much of what is known about synapses came from toxins and drugs that block one step at a time. For each one, find the step it blocks and then predict what the postsynaptic cell receives. Anything that blocks steps 1 to 8 gives the postsynaptic cell less signal. Anything that blocks step 9 gives it more, because removal is the only step whose job is to end the signal.

Curare and botulinum toxin both paralyze skeletal muscle, and they can be told apart by measuring acetylcholine release. With curare, acetylcholine is released in the normal amount, and it is the receptors that are occupied. With botulinum toxin no acetylcholine is released. Blocking acetylcholinesterase helps against curare, because curare competes with acetylcholine for the receptor and more acetylcholine displaces it. It cannot help against botulinum toxin, because there is no acetylcholine to preserve.

One honest note about SSRIs. Blocking the transporter raises serotonin in the cleft within hours, yet the improvement in mood usually takes several weeks. So the longer term changes that neurons make in response to the extra serotonin must matter too, and those are still being worked out. For this course, know the step that the drug blocks.

Worked problem 4 compares botulinum toxin with an organophosphate. They block opposite steps at the same synapses, and most of their signs are opposite. The exception is skeletal muscle, where both end in weakness.

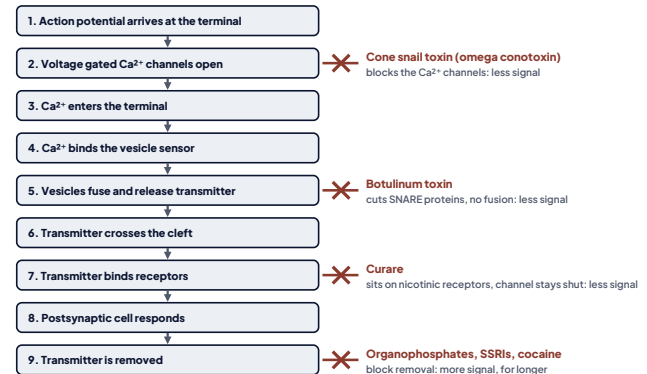


Figure 10. The nine steps from top to bottom, with the step each agent blocks.

SIX AGENTS, THE STEP EACH ONE BLOCKS, AND THE SIGNAL THE POSTSYNAPTIC CELL RECEIVES

AGENT	WHAT IT DOES	STEP	SIGNAL
Omega conotoxin, from cone snail venom	Blocks the voltage gated Ca^{2+} channels of the terminal	2	Less
Botulinum toxin	Cuts SNARE proteins, so vesicles cannot fuse	5	Less
Curare	Occupies nicotinic receptors without opening them	7	Less
Organophosphate insecticides and nerve agents	Inhibit acetylcholinesterase	9	More, for longer
SSRIs, such as fluoxetine and sertraline	Block the serotonin reuptake transporter	9	More, for longer
Cocaine	Blocks the dopamine reuptake transporter	9	More, for longer

WORKED PROBLEM 4

Botulinum toxin blocks release. An organophosphate blocks removal. Both act at every synapse that uses acetylcholine. Explain the signs of each at skeletal muscle, where the receptors are nicotinic, and at glands, gut, pupils and heart, where they are muscarinic, and explain why both end in muscle weakness.

Work out the acetylcholine in the cleft first. Then ask what each kind of target does with no signal, and with a signal that never ends.

THE WORK

- 1** **Botulinum toxin** acts at step 5. No vesicles fuse, so the acetylcholine in the cleft falls toward zero at every synapse that releases acetylcholine.
- 2** Skeletal muscle (nicotinic receptors) receives no signal: the muscles are weak and floppy. This is called flaccid paralysis. Glands and gut (muscarinic receptors) receive no signal either: dry mouth, dry eyes, constipation, and pupils that are wide and slow to react.
- 3** **An organophosphate** acts at step 9. Acetylcholinesterase is inhibited, so acetylcholine stays in the cleft and keeps binding at every one of the same synapses.
- 4** Glands, gut, airways and heart (muscarinic receptors) are stimulated continuously: saliva, tears, sweat and airway secretions increase, the gut is overactive, the airways narrow, the pupils become very small, and the heart rate slows.
- 5** Skeletal muscle (nicotinic receptors) is stimulated continuously too. At first the muscles twitch. Then they become weak. The reason is in the action potential lecture: a membrane that is held depolarized keeps its voltage gated sodium channels inactivated, so the muscle fiber can no longer fire.
- 6** So the two poisons give opposite signs at glands, gut, pupils and heart, and yet both can stop breathing, because a skeletal muscle needs a signal that turns on and then turns off.

The patient, explained

SLIDE 19 OF THE DECK

The baby has infant botulism. Honey can carry spores of the bacterium *Clostridium botulinum*. An adult's gut bacteria keep the spores from growing. An infant's gut bacteria are not yet established, so the spores grew in her large intestine and the bacteria made botulinum toxin. Honey is one known source of the spores. Soil and dust are another, and in many cases the source is never found.

The toxin was absorbed into her blood and taken up by axon terminals that release acetylcholine. It does not reach the brain in meaningful amounts, which is why she stayed alert and could feel. Inside the terminal, the toxin is an enzyme that cuts SNARE

proteins. Action potentials still arrived, Ca^{2+} channels still opened and Ca^{2+} still entered. The vesicles could not fuse, so no acetylcholine was released. Step 5 failed.

Every synapse that releases acetylcholine went quiet. The gut slowed first, which was the constipation. Then skeletal muscles stopped receiving a signal: the weak suck and cry, the drooping eyelids, the floppy neck, and last the diaphragm.

Treat it as an experiment. The variable that failed is the acetylcholine released per action potential. An antibody against the toxin binds toxin that is still in the blood, so it cannot enter any more terminals. It cannot reach toxin that is already inside terminals, so it stops the illness from progressing and does not reverse the weakness that is already there. A ventilator does the work of the diaphragm in the meantime. Recovery depends on the neurons themselves: they make new SNARE proteins and grow new terminal branches, which takes weeks to months. Infants who are supported through that time usually recover fully.

Compare this with organophosphate poisoning, where the opposite step fails. There, removal is blocked, acetylcholine stays in the cleft, and the signs at glands, gut, pupils and heart are the opposite ones.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

Proteins change shape when something binds to them. That is how a receptor or a channel opens.

An enzyme speeds up one reaction and is not used up by it.

Diffusion moves molecules from higher to lower concentration, and it is fast only over very short distances.

Chemistry review page

FROM EARLIER LECTURES

Resting potential lecture: E_{Ca} is about +123 mV, free Ca^{2+} is about 1 mM outside and 0.0001 mM inside, and ligand gated channels open when a molecule binds.

Neurons and glia lecture: the axon terminal is the output zone, fast axonal transport carries vesicles from the cell body, and astrocytes take up transmitter.

Action potential lectures: voltage gated channels, inactivation of sodium channels, graded potentials, and the frequency code.

Week 3: exocytosis, endocytosis and secondary active transport.

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

The Nernst equation with $z = +2$, which needs a calculator with a log button.

Time = distance ÷ speed, keeping the units the same.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

21

Sequence at a chemical synapse

Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

Do one of the two. Head your paper with the number.

21-1

Draw one axon terminal big enough to fill half your space, with the cleft and a patch of postsynaptic membrane. Number the events in order from the action potential arriving to the postsynaptic channel opening, one arrow per step, and show where calcium enters and what it does once inside. Then redraw the terminal sitting in a calcium free bath. One line: name the step where the chain stops, and say what still happens normally upstream of it.

21-2

Draw two traces stacked on one shared time axis in milliseconds, the presynaptic action potential on top and the postsynaptic membrane potential below. Mark the synaptic delay between them, about 0.5 ms. In the gap under that delay, draw the events that are happening while nothing shows on either trace. One line: which single step you think eats most of that delay, and why.

22**Neurotransmitter classes**

Group neurocrines into their seven chemical classes, acetylcholine, amines, amino acids, peptides, purines, gases, and lipids, and match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

Do one of the two. Head your paper with the number.

22-1

Draw a rough brain, a spinal cord, and one peripheral effector, side by side. Place acetylcholine, a catecholamine, serotonin, glutamate, GABA, glycine, and a neuropeptide at a real site of action on your drawing. Flag each one excitatory or inhibitory with an up or down arrow drawn on the postsynaptic membrane. One line: name the one you had to flag both ways, and say what forced that.

22-2

Draw acetylcholine arriving at two postsynaptic membranes side by side, a skeletal muscle fiber with nicotinic receptors and a cardiac pacemaker cell with muscarinic receptors. Show the ion moving at each and draw the membrane potential trace under each. Same transmitter, opposite result. One line: given both drawings, state what actually decides whether a synapse is excitatory or inhibitory.

23**Neurotransmitter removal**

Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

Do one of the two. Head your paper with the number.

23-1

Draw a synaptic cleft with transmitter still in it, then draw all three exit routes off that cleft, enzymatic breakdown, reuptake, and simple diffusion. Show where each one physically happens, on the postsynaptic side, on the presynaptic side, or out the edges. Name the transmitter you picked and put a real drug on the route it blocks. One line: what the postsynaptic cell experiences once that route is shut.

23-2

Draw the postsynaptic potential from one normal release event, membrane potential on the y axis and time on the x. On the same axes draw that trace again under two drugs, one that blocks degradation and one that blocks reuptake. Show what changes, the height, the width, or both. One line: which drug you expect to change the trace more, and what you would need to measure to be sure.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · The chemical synapse. Concentrations and voltages are typical teaching values. All figures are original drawings for this course.

Integration at the synapse

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The three worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 24 TO 28 OF WEEK 4

23 SLIDES · 3 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A single neuron in your spinal cord receives thousands of synapses. Some make it more likely to fire and some make it less likely, and at every moment the cell has one decision to make: send an action potential down its axon or stay silent. This lecture is about how that decision is made. We start with the two kinds of small potential a synapse can produce, the EPSP and the IPSP. Then we add them up at the trigger zone, with numbers. Last, we look at three ways the rules can be changed: inhibition of a single terminal, synapses that strengthen with use, and electrical synapses that use no transmitter at all.

Here is the problem we solve by the end. A gardener steps on a rusty nail. A week later his jaw is clamped shut, and the slightest noise or touch throws his whole body into painful spasms. He is awake through all of it. His muscles, and the motor neurons that drive them, are structurally normal. A bacterial toxin from the wound has reached his spinal cord.

Look at what the examination finds. He feels touch, sound and pain normally. He is awake and thinking clearly. His motor neurons carry action potentials, and when a motor neuron fires, its muscle contracts forcefully. Every part that excites a muscle is working. The one thing he has lost is the ability to keep muscles relaxed: a small stimulus makes opposing muscles contract together. Before you read further, write down a prediction. Is the toxin exciting his motor neurons, or is it removing something that normally holds them back?

See it move: will it fire?

SLIDE 13 OF THE DECK

This is a working model of the neuron in this lecture. It has four excitatory inputs, E1 to E4, at different distances along its

dendrites, one inhibitory input on a dendrite, I1, and one inhibitory input on the cell body, I2. The tracing on the right is the membrane potential at the trigger zone, with dashed lines at rest, -70 mV, and at threshold, -55 mV. Press the numbered cases in order, or choose which inputs fire yourself. The readouts beside the model give the highest point the trigger zone reached, how far that was from threshold, and whether the cell fired, and the sentence under the model explains why. Keep it open as you read and come back to it after each section.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-synaptic-integration-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-synaptic-integration-notes.html)

THREE THINGS TO TRY

- Press case 2, temporal summation. Then move the slider for the time between action potentials from 5 ms to 10 ms and to 25 ms. Write down the highest point each time and compare your numbers with worked problem 2. The model runs slightly higher than the hand calculation, because a real EPSP takes a moment to rise.
- Press case 3, spatial summation, and then slide E4 from 400 out to 800 micrometers. The cell stops firing, and nothing changed except where one synapse sits. That is case C of worked problem 3.
- Press case 4, which adds the inhibitory synapse on the cell body. Read the highest point. Then press Remove all inhibition and watch the same four EPSPs reach threshold. That is the patient in this lecture. Last, turn all the excitatory inputs off, leave I2 on, and move the chloride slider from 10 mM down to 5 mM to see the same synapse change from a small depolarization to a hyperpolarization.

Two kinds of postsynaptic potential

SLIDES 3 TO 5 OF THE DECK

The last lecture ended with one synapse: transmitter binds a receptor, a channel opens, and the postsynaptic membrane potential changes a little. That small change is a graded potential. From the graded potential lecture you know two things about it: it gets smaller as it spreads, and graded potentials can add. You also know that a neuron fires only if the membrane at its trigger zone, where the axon leaves the cell body, reaches threshold, about -55 mV. Synaptic integration is the name for the whole process in between: the neuron adds all of its excitatory and inhibitory inputs and gives one answer.

An excitatory postsynaptic potential, or EPSP, is a small depolarization. At most fast excitatory synapses in the brain, glutamate opens the AMPA receptor, which is a channel for small positive ions. Both Na^+ and K^+ can pass. At -70 mV the force on Na^+ is 131 mV inward, because E_{Na} is $+61$ mV, and the force on K^+ is only 25 mV outward, because E_{K} is -95 mV. So far more Na^+ enters than K^+ leaves, net positive charge comes in, and the membrane depolarizes. If the channel stayed open, the

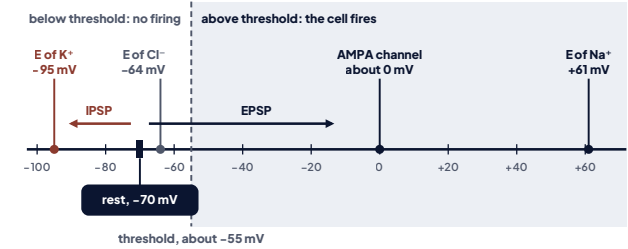
membrane would settle near 0 mV, which is well above threshold. That is what makes this synapse excitatory.

An inhibitory postsynaptic potential, or IPSP, makes the cell less likely to fire. There are two common kinds. One opens K^+ channels. K^+ leaves, and the membrane hyperpolarizes toward E_K , -95 mV. The GABA-B receptor does this through a G protein, so it is a metabotropic receptor from the last lecture, and its IPSP is slow. The other kind opens Cl^- channels. The GABA-A receptor and the glycine receptor are both Cl^- channels that open directly when the transmitter binds, so they are ionotropic and fast. Note that the transmitter does not decide whether a synapse excites or inhibits. The receptor and its channel decide. Acetylcholine excites skeletal muscle through one receptor and slows the heart through another.

Now the part that needs care. Use the rule from the resting potential lecture: opening channels for an ion moves the membrane toward that ion's equilibrium potential. With our concentrations, 110 mM chloride outside and 10 mM inside, E_{Cl} is -64 mV, which is a little above the resting potential of -70 mV. So opening Cl^- channels in our cell moves the membrane only slightly, and it moves it up, toward -64 mV. It is still inhibition, for two reasons. First, the open channels hold the membrane near -64 mV, which is 9 mV below threshold, and the membrane cannot be carried past -64 mV by chloride. Second, when an EPSP lifts the membrane above -64 mV, Cl^- enters through the open channels and cancels part of the positive charge that the EPSP brought in, so the EPSP is smaller than it would have been. This second effect is called shunting inhibition, because the open channels give the excitatory current a path out of the cell.

Cells and books vary on this point, and you should know why. Many adult neurons have a transporter that keeps Cl^- inside nearer 5 to 7 mM. With 110 mM outside, the Nernst equation then gives an E_{Cl} of about -83 to -74 mV, which is below rest, and opening Cl^- channels lets Cl^- in and hyperpolarizes the cell. Silverthorn describes the IPSP that way. Both descriptions are correct for the cells they describe. The general rule covers every case: a synapse is excitatory if its channel moves the membrane toward a value above threshold, and inhibitory if that value is below threshold.

Where each kind of channel moves the membrane, compared with threshold



A synapse is excitatory if its channel moves the membrane toward a value above threshold. It is inhibitory if that value is below threshold.

Figure 2. The equilibrium potentials from the first lecture this week, with threshold added.

Synaptic integration

The process by which a neuron adds all of its excitatory and inhibitory inputs and produces one output: fire or do not fire.

EPSP

Excitatory postsynaptic potential. A small, graded depolarization, usually from a cation channel such as the AMPA receptor letting Na^+ in.

IPSP

Inhibitory postsynaptic potential. A graded potential that makes firing less likely, from opening K^+ channels (hyperpolarizes toward -95 mV) or Cl^- channels (holds the membrane near E_{Cl}).

Shunting inhibition

Inhibition by open channels that let excitatory current leave the cell, so EPSPs are smaller even though the membrane potential itself hardly moved.

GABA-A and glycine receptors

Ionotropic receptors that are Cl^- channels. GABA is the main fast inhibitory transmitter in the brain and glycine in the spinal cord.

GABA-B receptor

A metabotropic receptor that opens K^+ channels through a G protein and gives a slow IPSP.

Working backwards from a recording

SLIDE 6 OF THE DECK

This problem is the reverse of the last section. You are given the recording and asked for the channel. It is worth doing carefully, because it is exactly what you will do in lab with the synapse simulation.

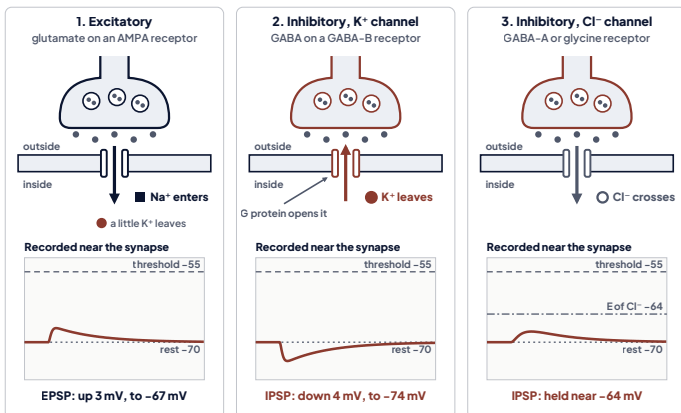


Figure 1. Three synapses and the potential each one produces. Sizes are illustrative. One synapse alone usually gives less than 1 mV.

WORKED PROBLEM 1

An electrode in a neuron records three traces on different trials. Each starts at -70 mV. The first rises 3 mV, the second falls 4 mV, and the third stays flat. For each, name the channel that opened and the ion that moved. Give the flat trace two different explanations, and say what second recording would tell them apart.

You need $E_K = -95$ mV, $E_{Cl} = -64$ mV with our numbers and lower in many neurons, the AMPA channel at about 0 mV, and $E_{Na} = +61$ mV.

THE WORK

- 1 The trace that rises 3 mV.** The membrane moved toward a value above -70 mV, and the channel that does this at a fast excitatory synapse is a cation channel such as the AMPA receptor. It passes Na^+ and K^+ . At -70 mV the force on Na^+ is 131 mV inward and the force on K^+ is only 25 mV outward, so far more Na^+ enters than K^+ leaves. Net positive charge enters, and the trace rises. This is an EPSP.
- 2 The trace that falls 4 mV.** The membrane moved toward a value below -70 mV. With our numbers only E_K , at -95 mV, is below rest, so a K^+ channel opened and K^+ left the cell. A GABA-B receptor does this through a G protein. In a neuron that keeps less chloride inside, a Cl^- channel could give the same trace, with Cl^- entering. Either way it is an IPSP.
- 3 The flat trace, explanation one.** Nothing happened at the synapse. The presynaptic neuron did not fire, or it fired and released no transmitter, so no channel opened.
- 4 The flat trace, explanation two.** A Cl^- channel opened in a cell whose chloride equilibrium potential is equal to its resting potential. There is no net force on Cl^- at that voltage, so no net charge crosses and the voltage does not move, even though the channels are open.
- 5 The second recording that separates them.** Fire a nearby excitatory input at the same moment and compare the EPSP with the EPSP you get from that input alone. If the quiet synapse opened no channels, the EPSP is its usual size. If it opened Cl^- channels, the EPSP is smaller, because as soon as the membrane rises above E_{Cl} , Cl^- enters and cancels part of the positive charge. That is how an input can be inhibitory while the membrane potential barely moves.

about 15 mV above rest. If nothing were lost on the way, it would take about 30 of those EPSPs arriving together to fire the cell. This is very different from the neuromuscular junction, where one action potential in the motor neuron always fires the muscle fiber. In the central nervous system no single synapse decides anything. A spinal motor neuron carries on the order of 10,000 synapses, and its output depends on many of them at once.

The arithmetic of integration can be written in one line. The potential at the trigger zone equals -70 mV, plus the EPSPs arriving, minus the IPSPs arriving, and the cell fires if that value reaches about -55 mV. Arriving means the size of each potential when it reaches the trigger zone, after it has shrunk with distance and with time. Adding potentials as plain numbers is a simplification that works well for small potentials. Real neurons add a little less than the arithmetic says, because each EPSP reduces the force that drives the next one.

There are two ways to add. In temporal summation, one presynaptic neuron fires several times in quick succession. Each EPSP begins before the one before it has decayed, so each starts from a higher level. In spatial summation, several presynaptic neurons fire at about the same moment and their EPSPs arrive at the trigger zone together. Real neurons use both at once: many inputs, each firing a train of action potentials. An EPSP and an IPSP that arrive together cancel, in whole or in part, in the same sum.

Temporal summation depends entirely on timing. After its peak, an EPSP falls by half roughly every 10 ms in the teaching values used here, because charge leaks back out through the membrane. If the next action potential arrives after 25 ms, less than a fifth of the first EPSP is left and almost nothing is added. If it arrives after 5 ms, about 71 percent is left. This is how the frequency code from the lecture on action potentials is read by the next cell. A faster train of action potentials in one axon produces a larger depolarization at the trigger zone of the cell it synapses on.

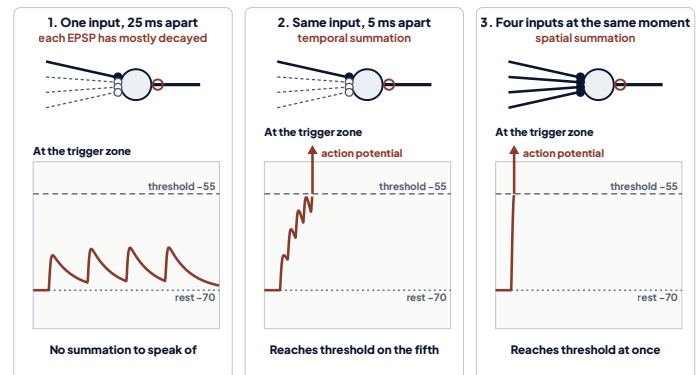


Figure 3. Tracings calculated with the same values as the moving model. Each EPSP is about 5 mV at the trigger zone.

One EPSP is small, so EPSPs have to add

SLIDES 7 TO 9 OF THE DECK

A single excitatory synapse on a neuron in the central nervous system typically produces an EPSP of roughly 0.5 mV at the cell body, and the value varies from synapse to synapse. Threshold is

WORKED PROBLEM 2

One excitatory input gives an EPSP of 5.4 mV at the trigger zone. After its peak, an EPSP falls by half about every 10 ms. The neuron rests at -70 mV and its threshold is -55 mV. Does the cell fire if this one input fires every 25 ms? Every 10 ms? Every 5 ms?

The peak after each new EPSP is 5.4 mV plus whatever is left of the earlier ones. What is left is one half after 10 ms, about 0.71 after 5 ms and about 0.18 after 25 ms.

THE WORK

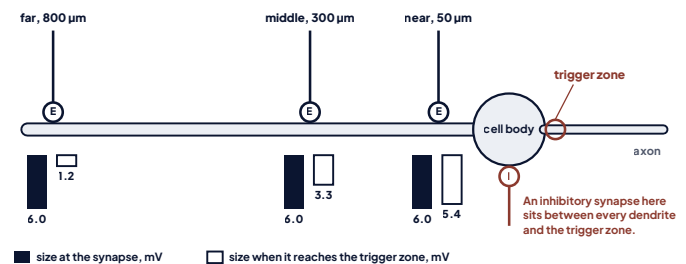
- 1 How far is threshold?** From -70 mV to -55 mV is 15 mV. One EPSP of 5.4 mV reaches -64.6 mV, which is 9.6 mV short.
- 2 Every 25 ms.** After 25 ms an EPSP has been halved two and a half times, so less than one fifth of it is left, about 1 mV. The second EPSP peaks near 6.4 mV above rest, and the later ones go no higher than about 6.6 mV. The cell does not fire however long the train continues.
- 3 Every 10 ms.** Exactly half is left each time. The peaks are 5.4, then $5.4 + 2.7 = 8.1$, then $5.4 + 4.05 = 9.5$, then 10.1 mV, and they level off below $5.4 \times 2 = 10.8$ mV. That is still 4 mV short of threshold.
- 4 Every 5 ms.** After 5 ms about 71 percent is left, because $0.71 \times 0.71 = 0.5$. The peaks are 5.4, 9.2, 11.9, 13.8 and then 15.2 mV. The fifth EPSP carries the trigger zone past 15 mV, and the cell fires.
- 5 What has to be true about timing.** The next EPSP has to arrive before the last one has decayed. The membrane potential takes roughly 10 ms to fall by half, so action potentials more than a few tens of milliseconds apart do not add, and action potentials a few milliseconds apart do. This is how the frequency code from the last lecture is read by the next cell: a faster train in one axon produces a larger depolarization at the trigger zone.

dendrite therefore has less effect at the trigger zone than the same synapse close to the cell body. In this lecture we use one teaching rule for that loss: the fraction that arrives is e raised to the power of minus the distance divided by 500 micrometers. With that rule, a 6 mV EPSP arrives as 5.4 mV from 50 micrometers, 3.3 mV from 300 micrometers and 1.2 mV from 800 micrometers. Real dendrites differ from one neuron to the next, and some have voltage gated channels that boost distant inputs, so treat the rule as an illustration of the direction of the effect.

Location matters for inhibition too. Inhibitory synapses are often found on the cell body and on the first part of the axon. Every EPSP from every dendrite has to pass that point on its way to the trigger zone, so one well placed inhibitory synapse can cancel the combined effect of many excitatory ones.

Two wiring patterns appear all through the nervous system. In divergence, the axon of one neuron branches and synapses on many target cells, so one signal reaches many places. In convergence, many presynaptic neurons synapse on one target cell, which integrates all of them. A spinal motor neuron is a good example of convergence: it receives sensory input, input from the brain and input from local inhibitory interneurons, and its one output decides whether a muscle contracts. The largest example in the body is the Purkinje cell of the cerebellum, which receives on the order of 100,000 to 200,000 synapses. That number is approximate.

The same 6 mV EPSP, started at three different distances from the cell body



Sizes use the teaching rule in this lecture: a graded potential shrinks by the factor $e^{-(\text{distance} / 500 \mu\text{m})}$.

Figure 4. Three identical synapses at different distances, and one inhibitory synapse on the cell body.

FOUR EXCITATORY INPUTS, EACH 6 MV AT ITS OWN SYNAPSE, AND ONE INHIBITORY INPUT

INPUT	DISTANCE FROM THE CELL BODY	FRACTION THAT ARRIVES	SIZE AT THE TRIGGER ZONE
E1	50 μm	0.90	+5.4 mV
E2	150 μm	0.74	+4.4 mV
E3	350 μm	0.50	+3.0 mV
E4	400 μm	0.45	+2.7 mV
I1 (-5 mV at its synapse)	250 μm	0.61	-3.0 mV

Temporal summation

Adding of graded potentials that arrive one after another from the same input, close enough in time that each begins before the last has decayed.

Spatial summation

Adding of graded potentials that arrive at the trigger zone at the same time from different synapses.

Trigger zone

The region where the axon leaves the cell body, rich in voltage gated Na^+ channels. It is where the sum of all inputs is compared with threshold.

Location, and a full worked problem

SLIDES 10 TO 12 OF THE DECK

A graded potential loses size as it spreads, because charge leaks out through the membrane along the way. A synapse far out on a

WORKED PROBLEM 3

Using the table above, with rest at -70 mV and threshold at -55 mV, decide whether the cell fires in each case. A: E1 alone. B: all four excitatory inputs together. C: the same four, with E4 moved out to 800 micrometers, where the fraction that arrives is 0.20. D: case B plus I1. E: case D, but the inhibitory terminal cannot release its transmitter.

Add the sizes at the trigger zone to -70 mV and compare the result with -55 mV.

THE WORK

- Case A, E1 alone.** $-70 + 5.4 = -64.6$ mV. That is 9.6 mV short of threshold, so the cell does not fire.
- Case B, all four excitatory inputs at the same moment.** $5.4 + 4.4 + 3.0 + 2.7 = 15.5$ mV. $-70 + 15.5 = -54.5$ mV. That is above -55 mV, so the cell fires. This is spatial summation.
- Case C, the same four, with E4 moved out to 800 μm .** E4 now arrives as $6 \times 0.20 = 1.2$ mV. The sum is $5.4 + 4.4 + 3.0 + 1.2 = 14.0$ mV, which gives -56.0 mV. The cell does not fire. Nothing about the synapse changed except where it sits.
- Case D, all four as in case B, plus I1.** I1 makes -5 mV at $250 \mu\text{m}$ and arrives as $-5 \times 0.61 = -3.0$ mV. The sum is $15.5 - 3.0 = 12.5$ mV, which gives -57.5 mV. The cell does not fire. One inhibitory input has canceled the effect of four excitatory ones.
- Case E, case D with the inhibitory terminal unable to release.** The -3.0 mV is gone, the sum is back to 15.5 mV, and the cell fires again. Keep this case in mind for the patient.
- The simplification.** Adding potentials as plain numbers is close to right for small potentials and is how these problems are worked in this course. In a real neuron the sum comes out a little smaller than the arithmetic, and an inhibitory synapse that opens Cl^- channels on the cell body removes more than its own size suggests. The model on slide 13 shows that case.

ONE MORE TO TRY ON YOUR OWN

A neuron has three excitatory inputs worth about 4 mV each at the trigger zone and two inhibitory inputs worth about 5 mV each. Can it ever reach threshold?

All three excitatory inputs fired once, at the same moment, give 12 mV, which is 3 mV short. So single volleys cannot do it, even with no inhibition. Temporal summation can. If all three fire twice, 5 ms apart, the second volley adds 12 mV to the $0.71 \times 12 = 8.5$ mV that is left, which is 20.5 mV and well past threshold. Add one inhibitory input at the time of the second volley and the total falls to 15.5 mV, which still fires. Add both and it falls to 10.5 mV, which does not.

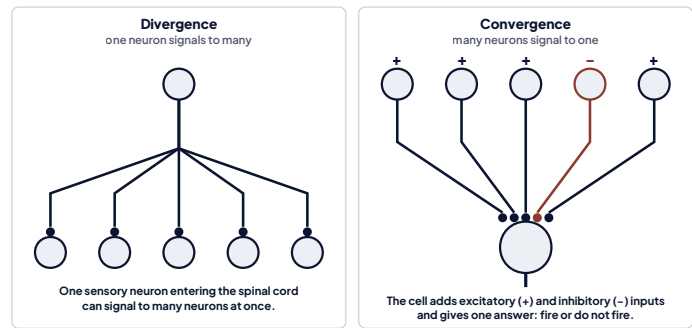


Figure 5. Divergence and convergence.

Divergence

One neuron synapsing on many target cells.

Convergence

Many neurons synapsing on one target cell, which integrates them.

Presynaptic inhibition and facilitation

SLIDES 14 TO 15 OF THE DECK

Everything so far has acted on the receiving cell. There is another place to act. In presynaptic inhibition a modulatory neuron ends on the axon terminal of another neuron, which is called an axoaxonic synapse. Its transmitter reduces the amount of Ca^{2+} that enters that terminal when an action potential arrives. From the last lecture you know that Ca^{2+} entry is what makes vesicles fuse. Less Ca^{2+} entering means fewer vesicles fuse, less transmitter is released, and the EPSP from that one input is smaller. The postsynaptic cell has not changed at all, so it responds to its other inputs exactly as before.

Compare that with postsynaptic inhibition, which is the IPSP from the first section. It acts on the receiving cell, so it reduces the effect of every input to that cell. The advantage of presynaptic control is selectivity: one input can be turned down while the others are left alone. If you needed to cut the effect of input A by half and leave inputs B and C untouched, an inhibitory synapse on the terminal of A would do it. An inhibitory synapse on the cell body would fail the requirement, because it would lower B and C by just as much.

Presynaptic facilitation is the opposite change at a terminal. The modulatory transmitter keeps the terminal depolarized a little longer when an action potential arrives, more Ca^{2+} enters, and more transmitter is released. It was worked out in the sea slug *Aplysia*, where serotonin closes K^+ channels in a sensory terminal.

The standard example of presynaptic inhibition in the human body is pain. Pain fibers make their first synapse in the dorsal horn of the spinal cord. Interneurons there release enkephalins, the body's own opioids, onto the terminals of the pain fibers. Opioid receptors on those terminals reduce Ca^{2+} entry, less transmitter is released, and less of the pain signal reaches the next neuron. Morphine and the other opioid drugs act at the same receptors. Opioids also act at other sites, including on the postsynaptic cell and in the brain, so this is one part of how they work and not the whole of it.

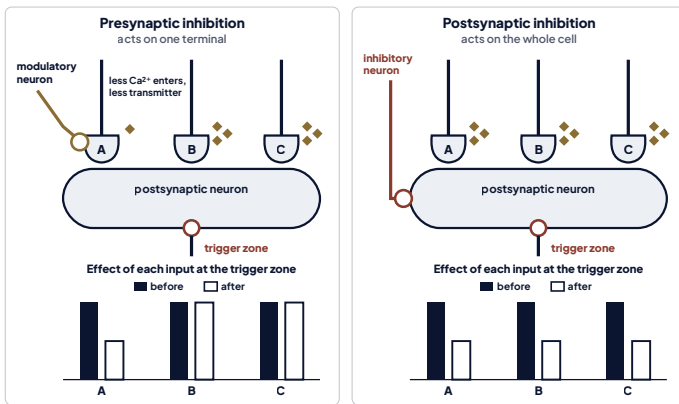


Figure 6. The same three inputs, inhibited in two different places.

Axooxonic synapse

A synapse made by one axon terminal onto another axon terminal.

Presynaptic inhibition

A decrease in transmitter release from one terminal, caused by a modulatory input that reduces Ca^{2+} entry into that terminal. It is selective for that one input.

Presynaptic facilitation

An increase in transmitter release from one terminal, caused by a modulatory input that increases Ca^{2+} entry.

Postsynaptic inhibition

An IPSP in the receiving cell. It reduces the effect of all inputs to that cell.

Synaptic plasticity

SLIDES 16 TO 18 OF THE DECK

Synaptic plasticity means that the strength of a synapse depends on its history. The same presynaptic action potential can produce a larger or a smaller EPSP than it did before. Some of these changes are brief. In facilitation, each EPSP in a short rapid train is larger than the one before, because Ca^{2+} from one action potential has not been cleared from the terminal when the next arrives, so more vesicles fuse. In depression, the EPSPs get smaller during a long rapid train, because the terminal is using vesicles faster than it can refill them. Both changes are in the presynaptic terminal, and both fade within seconds to minutes after the train stops. A memory lasts far longer than that, so there must also be changes that last.

Long term potentiation, or LTP, is an increase in synaptic strength that lasts hours or longer. It was discovered in the hippocampus, a part of the brain needed for forming new memories, and it is best understood at glutamate synapses. Take it one step at a time. Glutamate binds two receptors on the postsynaptic membrane. The AMPA receptor opens at once and produces the ordinary EPSP. The NMDA receptor also binds glutamate, but near the resting potential its channel is plugged by a Mg^{2+} ion from the outside fluid, which the negative interior holds in the pore. So after one action potential, the NMDA receptor has glutamate on it and passes almost nothing.

A rapid burst of input changes that. Many EPSPs add, by the temporal and spatial summation you have already worked through, and the membrane depolarizes strongly. A less negative interior no longer holds the positive Mg^{2+} ion, so the plug leaves

the pore. With glutamate still bound, the NMDA channel is now open, and it lets Ca^{2+} in as well as Na^{+} . Ca^{2+} acts as a second messenger. It activates protein kinases in the postsynaptic cell, and these add more AMPA receptors to the membrane and make the existing ones pass more current. The postsynaptic cell may also send a signal back across the cleft so that the terminal releases more glutamate. Changes that last for days also need new proteins to be made.

The result is that the same single action potential now produces a larger EPSP. Notice why one action potential alone cannot do this. The NMDA receptor opens only when two things are true together: glutamate is bound, which says the presynaptic cell fired, and the membrane is already strongly depolarized, which says the postsynaptic cell was strongly driven at that moment. One EPSP of half a millivolt does not remove the Mg^{2+} .

In the standard experiment, the EPSP is tested with a single action potential once a minute. After one burst of rapid stimulation it jumps, settles at a larger size than before and stays there for as long as the recording lasts. If NMDA receptors are blocked with a drug, ordinary transmission is normal, because AMPA receptors carry it. The burst still causes a brief rise, which is short term plasticity in the terminal, but the lasting increase does not happen. So the AMPA receptor carries the ordinary, moment to moment signal, and the NMDA receptor carries the signal that something should change.

Long term depression, or LTD, is the reverse. Prolonged slow stimulation lets in a small amount of Ca^{2+} , which activates a different set of enzymes. AMPA receptors are removed from the membrane, and the EPSP becomes smaller for a long time. Most neuroscientists think changes like these are part of how learning and memory work, and blocking NMDA receptors in the hippocampus does impair some kinds of learning in animals. How a particular memory is stored is still being worked out, so treat LTP as a well supported mechanism for strengthening a synapse and not as a full explanation of memory.

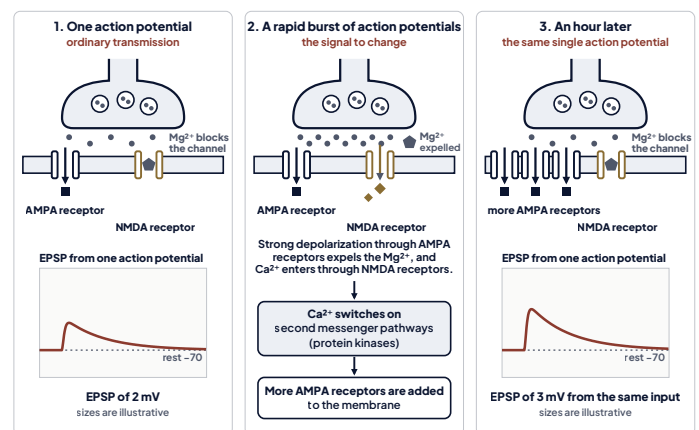


Figure 7. One glutamate synapse before, during and after a rapid burst of activity.

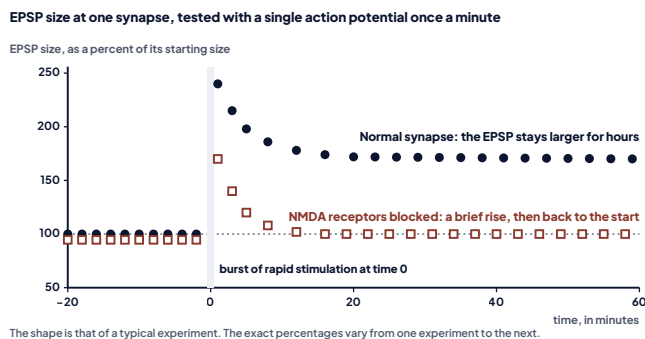


Figure 8. The size of the EPSP before and after one burst of rapid stimulation, with and without working NMDA receptors.

Synaptic plasticity

A change in the strength of a synapse that depends on its earlier activity.

Facilitation and depression

Short term increases and decreases in transmitter release during a train of action potentials, lasting milliseconds to minutes.

AMPA receptor

An ionotropic glutamate receptor that opens at once and produces the ordinary fast EPSP.

NMDA receptor

An ionotropic glutamate receptor whose channel is blocked by Mg^{2+} near rest. It opens only when glutamate is bound and the membrane is strongly depolarized, and it lets Ca^{2+} in.

Long term potentiation (LTP)

A long lasting increase in the size of the EPSP at a synapse after strong, rapid activity. It depends on Ca^{2+} entry through NMDA receptors and on adding AMPA receptors.

Long term depression (LTD)

A long lasting decrease in the size of the EPSP, with AMPA receptors removed from the membrane.

Electrical synapses

SLIDES 19 TO 20 OF THE DECK

Not every synapse uses a transmitter. At an electrical synapse two cells are joined by gap junctions. Each gap junction channel is built from proteins called connexins. Six connexins form a half channel in one membrane, six more form a half channel in the other, and the two line up to make one pore from the cytoplasm of one cell to the cytoplasm of the next. Where they occur, the two membranes lie only about 3.5 nm apart, compared with a cleft of about 20 to 40 nm at a chemical synapse.

Ions carry current straight through the pore, so a depolarization in one cell depolarizes its neighbor with almost no delay. A chemical synapse takes about 0.5 to 1 ms, and sometimes longer, because Ca^{2+} has to enter, vesicles have to fuse, and transmitter has to cross and bind. Current can usually pass through a gap junction in either direction, so a group of cells joined this way tends to fire together. You find electrical synapses in heart muscle, in the smooth muscle of the gut and the uterus, between glial cells, and between some neurons in the brain.

What an electrical synapse gives up is flexibility. The signal keeps its sign, so a depolarization can only pass on as a smaller

depolarization. It cannot be turned into inhibition, it cannot be amplified, and there is no terminal to modulate and no receptor number to change. Gap junctions can open and close, but that is a much coarser control. So the two kinds of synapse suit different jobs. When every muscle cell in the wall of the heart must contract at nearly the same moment, gap junctions are the right choice, because delays at millions of chemical synapses would add up and the contraction would not be coordinated. When a spinal motor neuron must be able to ignore a sensory signal on some occasions and respond to it on others, only a chemical synapse will do. The delay of a chemical synapse is the cost of being able to integrate.

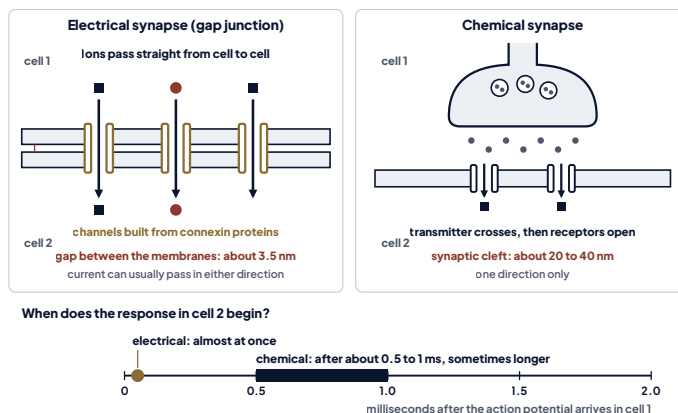


Figure 9. The two kinds of synapse, with the time each one takes.

TWO WAYS FOR ONE CELL TO SIGNAL TO THE NEXT

	CHEMICAL SYNAPSE	ELECTRICAL SYNAPSE
What crosses	Neurotransmitter, across a cleft of about 20 to 40 nm	Ions, through connexin channels across a gap of about 3.5 nm
Delay	About 0.5 to 1 ms, sometimes longer	Almost none
Direction	One way	Usually both ways
Sign of the signal	Excitatory or inhibitory, depending on the receptor	The same sign in both cells
Amplification	Yes	No. The signal can only get smaller
Adjustable	Highly: presynaptic control, receptor number, plasticity	Much less
Best suited for	Integration and decisions	Speed and firing together

Electrical synapse

A junction where current passes directly from one cell to the next through gap junction channels.

Gap junction

A cluster of channels that join the cytoplasm of two cells. Ions and small molecules pass through.

Connexin

The protein that gap junction channels are built from, six in each membrane.

The patient, explained

SLIDES 21 TO 22 OF THE DECK

Start with a normal spinal motor neuron. It receives excitatory input all the time, from sensory neurons and from the brain. Inhibitory interneurons that release glycine or GABA keep it below threshold unless the excitation is strong. Put illustrative numbers on it. An excitatory input worth 18 mV and an inhibitory input worth -8 mV arrive together, and $-70 + 18 - 8 = -60$ mV, so the cell does not fire. Now block release from the inhibitory terminal. $-70 + 18 = -52$ mV, which is above threshold. The same input that did nothing a moment ago now fires the motor neuron.

That is what happened to the gardener. The bacterium *Clostridium tetani* grew in the deep wound and released tetanus toxin. Motor axon terminals took the toxin up, and retrograde axonal transport, from the neuron lecture, carried it back to the cell bodies in the spinal cord and brainstem. That trip is part of why a week went by. In the spinal cord the toxin passed from the motor neuron into the terminals of the inhibitory interneurons that synapse on it. There it cut one of the proteins that a vesicle needs in order to fuse with the membrane, so those terminals could no longer release glycine or GABA.

Nothing was added to the excitatory side. His motor neurons still receive the same EPSPs from every touch and sound, but the inhibitory term is gone from the sum, so ordinary input now reaches threshold. Normally, when one muscle contracts, inhibitory interneurons keep the opposing muscle quiet. Without them, both contract at once and with full force. The strong jaw closing muscles clamp the jaw shut, which gives the disease its old name, lockjaw, and any stimulus sets off a spasm of the whole body.

Treat it as an experiment. The variable that failed is inhibitory transmitter release onto motor neurons, and through it the sum at the trigger zone. Three physiological targets follow from that. The GABA-A receptors on the motor neuron are intact, so a drug that makes those channels open more readily, such as a benzodiazepine, increases whatever inhibition remains and puts a negative term back into the sum. Every sound and touch adds EPSPs, so a quiet, dark room lowers the excitatory side of the same sum. And antibody against the toxin binds whatever toxin has not yet entered neurons. Toxin already inside the terminals cannot be reached, so recovery waits for those terminals to be repaired or replaced, which takes weeks. Vaccination works by having that antibody present before any toxin arrives.

Two comparisons test whether you have the mechanism. Strychnine blocks the glycine receptor on the motor neuron. Glycine is still released, but its receptor cannot open its Cl^- channel, so there is no IPSP, and the result is convulsions set off by small stimuli, very much like tetanus. One agent acts before the cleft and the other after it, and both remove the same term from the sum. Botulinum toxin is a close relative of tetanus toxin and cuts the same family of vesicle proteins, but it stays in the motor neuron terminal at the muscle, where it blocks acetylcholine release. The muscle cannot be excited at all, and the result is limp paralysis. The mechanism is the same, the synapse is different, and the effect is the opposite.

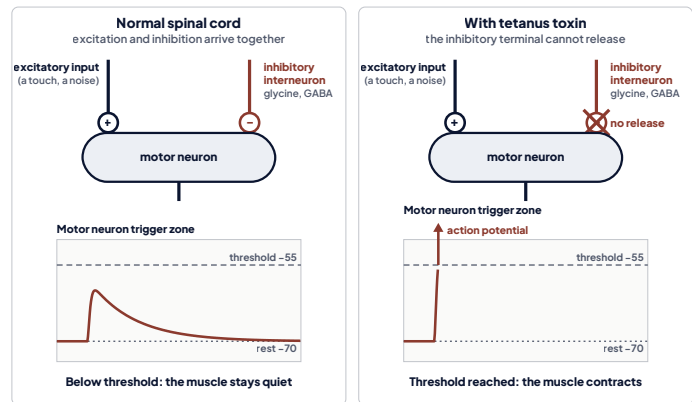


Figure 10. A spinal motor neuron with the same excitatory input in both panels.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

Ions and charge: Na^+ , K^+ , Cl^- , Ca^{2+} and Mg^{2+} , and that like charges repel.

Proteins as channels, receptors and enzymes, and what a protein kinase does.

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential: $E_K -95$ mV, $E_{\text{Na}} +61$ mV, $E_{\text{Cl}} -64$ mV, rest -70 mV, and the rule that opening channels for an ion moves the membrane toward that ion's equilibrium potential.

Neurons and glia: dendrites and cell body as the input zone, the trigger zone, and retrograde axonal transport.

Graded potentials: they shrink with distance and can add, and threshold is about -55 mV.

How action potentials carry information: the frequency code.

The chemical synapse: Ca^{2+} entry and vesicle fusion, ionotropic and metabotropic receptors, glutamate, GABA and glycine.

The only anatomy used is the dorsal horn of the spinal cord, where pain fibers make their first synapse, and the hippocampus and cerebellum, which are named and not described.

MATH

Adding and subtracting signed numbers.

Multiplying by a fraction, for example 6×0.45 .

No logarithms are needed in this lecture. The one Nernst value, E_{Cl} , is given.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Competency 29, the synapse simulation, is done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

24

Excitatory and inhibitory postsynaptic potentials

Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change, and contrast fast responses through ionotropic receptor-channels with slow responses through metabotropic G protein coupled receptors, including how a neuromodulator differs from a neurotransmitter.

Do one of the two. Head your paper with the number.

24-1

Draw a postsynaptic neuron with two synapses on one dendrite, one excitatory and one inhibitory. At each, draw the channel that opened and the ion moving through it with an arrow for direction. Under each synapse draw its own trace starting from about -70 mV and label the peak value. One line: how an inhibitory input can still be inhibitory even when the membrane potential barely moves.

24-2

Draw three recorded traces from the same neuron on shared axes, all starting near -70 mV, one rising 3 mV, one falling 4 mV, and one staying flat. For each trace, work backwards and draw the channel and the ion that would produce it. Give the flat trace two different explanations. One line: which explanation a second recording could rule out, and what that recording would be.

25

Summation and integration

Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

Do one of the two. Head your paper with the number.

25-1

Draw the same neuron twice. On the first, one presynaptic neuron fires four times in quick succession. On the second, four separate presynaptic neurons fire at the same moment. Under each, draw the trace at the trigger zone with a resting line near -70 mV and a dashed threshold line near -55 mV. Label which is temporal and which is spatial. One line: what has to be true about timing for the first one to work at all.

25-2

Draw a neuron with five inputs, three excitatory worth about 4 mV each and two inhibitory worth about 5 mV each. Choose which fire and in what order, then draw the trace at the trigger zone against a threshold line near -55 mV, writing your running total along the trace. Now change one input, its timing or its sign, so the outcome flips. Write which change you made and why it was enough.

26

Presynaptic modulation

Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

Do one of the two. Head your paper with the number.

26-1

Draw a neuron with three excitatory inputs converging on it. Add a modulatory neuron that synapses on one input's terminal, not on the cell body. Show what it does to calcium entry and to transmitter release at that terminal only. Draw the postsynaptic response to all three inputs before and after. One line: what postsynaptic inhibition would have done to the other two inputs.

26-2

You need to cut one input's effect by half and leave the other inputs untouched. Draw two attempts side by side, one acting on that input's presynaptic terminal and one acting on the postsynaptic membrane. On each drawing, shade every synapse the attempt affected. Mark the attempt that failed the requirement. One line: name the property that makes presynaptic control the right tool for this job.

27

Synaptic plasticity

Explain long term potentiation as a mechanism of learning, tracing how glutamate acts on AMPA and NMDA receptors, how strong depolarization removes the magnesium block, how calcium entry starts second messenger pathways, and how repeated stimulation leads to insertion of new AMPA receptors.

Do one of the two. Head your paper with the number.

27-1

Draw a glutamate synapse with both AMPA and NMDA receptors in the postsynaptic membrane. Show the magnesium sitting in the NMDA channel at rest. In a second panel, draw the same synapse during repeated high frequency stimulation with the magnesium displaced and calcium entering. In a third, draw what the membrane looks like an hour later. One line: why one presynaptic action potential on its own cannot do this.

27-2

Draw a graph of postsynaptic response size against time, with a burst of high frequency stimulation marked on the time axis, and plot the response before, during, and long after the burst. On the same axes, draw a second curve for the identical experiment with NMDA receptors blocked. One line: which receptor carries the ordinary moment to moment signal, and which one carries the signal that something should change.

28

Electrical synapses

Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

Do one of the two. Head your paper with the number.

28-1

Draw two synapses side by side in cross section at the same scale, an electrical one at a gap junction and a chemical one. Show what physically crosses at each and make clear which gap is wider. Underneath both, draw one shared time axis and mark when the postsynaptic response begins for each. One line: what the chemical synapse buys with the delay it pays.

28-2

Pick two real jobs, one where many cells must fire as a single unit and one where a signal must be turned up, turned down, or blocked outright. Draw the synapse you would build for each and label it electrical or chemical. On each drawing, circle the feature that made it right for that job. One line: what fails if you swap the two.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Integration at the synapse. Potential sizes and distances are teaching values. All figures are original drawings for this course.